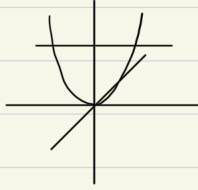


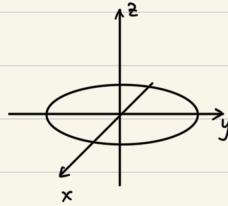
数学分析 B2 期末复习.

一、三重积分

$$\begin{aligned}
 1. \iiint_V (x^2 + y^2) dx dy dz &= \int_0^8 \left(\iint_{x^2+y^2 \leq 2z} (x^2 + y^2) dx dy \right) dz \\
 &= \int_0^8 \left(\int_0^{\sqrt{2z}} \int_0^{\sqrt{2z-r^2}} r^2 \cdot r dr d\varphi \right) dz \\
 &= \int_0^8 2\pi \cdot \frac{(\sqrt{2z})^4}{4} dz \\
 &= \int_0^8 2\pi z^2 dz \\
 &= \frac{2\pi}{3} \cdot 8^3 \\
 &= \frac{1024\pi}{3}
 \end{aligned}$$

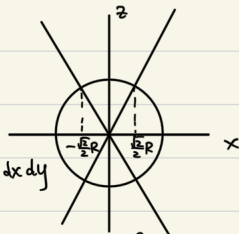


$$\begin{aligned}
 2. \iiint_V (x+y+z) dx dy dz &= I_1 + I_2 + I_3 \\
 I_1 &= \iiint_V x dx dy dz = 0 \\
 I_2 &= \iiint_V y dx dy dz = 0 \\
 I_3 &= \iiint_V z dx dy dz = \int_0^c \left[\iint_{\frac{x^2}{a^2} + \frac{y^2}{b^2} \leq 1 - \frac{z^2}{c^2}} dx dy \right] z dz \\
 &= \int_0^c \pi \cdot ab \left(1 - \frac{z^2}{c^2}\right) z dz \\
 &= \pi ab \left(\frac{c^2}{2} - \frac{1}{2} \frac{c^4}{c^2}\right) \\
 &= \frac{\pi}{4} abc^2
 \end{aligned}$$



3. 方法一:

$$\begin{aligned}
 &\iiint_V \sqrt{x^2 + y^2 + z^2} dx dy dz \\
 &= \iint_{x^2+y^2 \leq (\frac{5}{2}R)^2} \left(\int_{\sqrt{x^2+y^2}}^{\sqrt{R^2-x^2-y^2}} \sqrt{x^2+y^2+z^2} dz \right) dx dy \\
 &\int \frac{1}{\sqrt{a^2+z^2}} dz = \frac{z = a \tan \theta}{\cos \theta} \int \frac{1}{\cos \theta} d\theta = \int \sec \theta d\theta = \ln |\sec \theta + \tan \theta| \\
 &= \ln |\sec \theta + \tan \theta| = \ln \left| \sqrt{\frac{z^2}{a^2} + 1} + \frac{z}{a} \right| + C
 \end{aligned}$$



$$\begin{aligned}
 \int \sqrt{a^2+z^2} dz &= z \sqrt{a^2+z^2} - \int \frac{z^2}{\sqrt{a^2+z^2}} dz \\
 &= z \sqrt{a^2+z^2} - \int \frac{a^2+z^2-a^2}{\sqrt{a^2+z^2}} dz \\
 &= z \sqrt{a^2+z^2} - \int \sqrt{a^2+z^2} dz + a^2 \int \frac{1}{\sqrt{a^2+z^2}} dz
 \end{aligned}$$

$$\Rightarrow \int \sqrt{a^2+z^2} dz = \frac{1}{2} z \sqrt{a^2+z^2} + \frac{a^2}{2} \ln |z + \sqrt{z^2+a^2}| + C$$

$$\begin{aligned}
 \int_{\sqrt{x^2+y^2}}^{\sqrt{R^2-x^2-y^2}} \sqrt{x^2+y^2+z^2} dz &= \left(\frac{1}{2} \sqrt{R^2-x^2-y^2} \cdot R + \frac{x^2+y^2}{2} \ln \left(\sqrt{R^2-x^2-y^2} + R \right) \right) \\
 &\quad - \left(\frac{1}{2} \sqrt{x^2+y^2} \cdot \sqrt{2} \sqrt{x^2+y^2} + \frac{x^2+y^2}{2} \ln (1 + \sqrt{2}) \sqrt{x^2+y^2} \right)
 \end{aligned}$$

$$\begin{aligned}
 \iiint_V \sqrt{x^2+y^2+z^2} dx dy dz &= \iint_{r \leq \frac{\sqrt{2}}{2} R} \left(\frac{1}{2} R \sqrt{R^2-r^2} - \frac{\sqrt{2}}{2} r^2 + \frac{r^2}{2} \ln \frac{\sqrt{R^2-r^2}+R}{(1+\sqrt{2})r} \right) r dr d\varphi \\
 &= 2\pi \int_0^{\frac{\sqrt{2}}{2} R} \left(\frac{1}{2} R \sqrt{R^2-r^2} - \frac{\sqrt{2}}{2} r^2 + \frac{r^2}{2} \ln \frac{\sqrt{R^2-r^2}+R}{(1+\sqrt{2})r} \right) r dr
 \end{aligned}$$

$$\begin{aligned}
 I_1 &= \int_0^{\frac{\sqrt{2}}{2} R} \frac{1}{2} R \sqrt{R^2-r^2} \cdot r dr = \frac{R}{4} \int_{\frac{1}{2} R^2}^{R^2} \sqrt{u} du = \frac{R}{4} \left. \frac{u^{\frac{3}{2}}}{\frac{3}{2}} \right|_{\frac{1}{2} R^2}^{R^2} \\
 &= \frac{R}{6} \cdot \left(R^3 - \frac{1}{2\sqrt{2}} R^3 \right) \\
 &= \frac{R^4}{6} \left(1 - \frac{1}{2\sqrt{2}} \right)
 \end{aligned}$$

$$I_2 = \int_0^{\frac{\sqrt{2}}{2} R} -\frac{\sqrt{2}}{2} r^2 \cdot r dr = -\frac{\sqrt{2}}{2} \frac{r^4}{4} \Big|_0^{\frac{\sqrt{2}}{2} R} = -\frac{\sqrt{2}}{8} \frac{R^4}{4} = -\frac{\sqrt{2} R^4}{32}$$

$$I_3 = \int_0^{\frac{\sqrt{2}}{2} R} \frac{r^2}{2} \ln \frac{\sqrt{R^2-r^2}+R}{(1+\sqrt{2})r} \cdot r dr = \frac{1}{2} R^4 \int_0^{\frac{\sqrt{2}}{2}} t^3 \ln \frac{\sqrt{1-t^2}+1}{(1+\sqrt{2})t} dt$$

$$\begin{aligned}
 A &= \int_0^{\frac{\sqrt{2}}{2}} t^3 \ln (\sqrt{1-t^2}+1) dt \stackrel{u=\sqrt{1-t^2}}{=} \int_{\frac{\sqrt{2}}{2}}^1 (1-u^2) \ln (u+1) u du \\
 &= \left(\frac{u^2}{2} - \frac{u^4}{4} \right) \ln (u+1) \Big|_{\frac{\sqrt{2}}{2}}^1 - \int_{\frac{\sqrt{2}}{2}}^1 \frac{\frac{u^2}{2} - \frac{u^4}{4}}{u+1} du \\
 &= \frac{1}{4} \ln 2 - \frac{3}{16} \ln (\frac{\sqrt{2}}{2}+1) - \frac{1}{4} \int_{\frac{\sqrt{2}}{2}}^1 \left(-u^3 + u^2 + u - 1 + \frac{1}{u+1} \right) du \\
 &= \frac{1}{4} \ln 2 - \frac{3}{16} \ln (\sqrt{2}+1) + \frac{3}{32} \ln 2 + \frac{1}{4} \left(\frac{1}{4} - \frac{1}{3} - \frac{1}{2} + 1 - \ln 2 \right) \\
 &\quad - \frac{1}{16} + \frac{\sqrt{2}}{12} + \frac{1}{4} - \frac{\sqrt{2}}{2} + \ln (\sqrt{2}+1) - \frac{1}{2} \ln 2 \\
 &= \frac{29}{192} - \frac{5\sqrt{2}}{48} + \frac{1}{16} \ln (\sqrt{2}+1) - \frac{1}{32} \ln 2
 \end{aligned}$$

$$B = \int_0^{\frac{\sqrt{2}}{2}} -t^3 \ln (1+\sqrt{2}) dt = -\frac{1}{16} \ln (1+\sqrt{2})$$

$$C = \int_0^{\frac{\sqrt{2}}{2}} -t^3 \ln t dt = -\frac{t^4}{4} \ln t \Big|_0^{\frac{\sqrt{2}}{2}} + \frac{t^4}{16} \Big|_0^{\frac{\sqrt{2}}{2}} = \frac{1}{32} \ln 2 + \frac{1}{64}$$

$$I_3 = \frac{1}{2} R^4 \left(\frac{29}{192} - \frac{5\sqrt{2}}{48} + \frac{1}{64} \right) = \frac{1}{12} R^4 - \frac{5\sqrt{2}}{96} R^4$$

$$\begin{aligned}
 I_1 + I_2 + I_3 &= \frac{R^4}{6} \left(1 - \frac{1}{2\sqrt{2}} \right) - \frac{\sqrt{2} R^4}{32} + \frac{1}{12} R^4 - \frac{5\sqrt{2}}{96} R^4 \\
 &= \frac{R^4}{4} - \frac{\sqrt{2} R^4}{8}
 \end{aligned}$$

$$\begin{aligned}
 \iiint_V \sqrt{x^2+y^2+z^2} dx dy dz &= 2\pi (I_1 + I_2 + I_3) \\
 &= \frac{\pi R^4}{4} (2 - \sqrt{2})
 \end{aligned}$$

方法二: 球坐标换元

$$x = \rho \sin \theta \cos \phi$$

$$y = \rho \sin \theta \sin \phi$$

$$z = \rho \cos \theta$$

$$0 \leq \theta \leq \frac{\pi}{4}, 0 \leq \phi \leq 2\pi, 0 \leq \rho \leq R.$$

$$\begin{aligned}
 \iiint_V \sqrt{x^2+y^2+z^2} dx dy dz &= \int_0^R \int_0^{\frac{\pi}{4}} \int_0^{2\pi} \rho \cdot \rho^2 \sin \theta d\phi d\theta d\rho \\
 &= \int_0^R \rho^3 d\rho \int_0^{\frac{\pi}{4}} \sin \theta d\theta \int_0^{2\pi} d\phi \\
 &= \frac{R^4}{4} \cdot \left(1 - \frac{\sqrt{2}}{2} \right) \cdot 2\pi \\
 &= \frac{\pi R^4}{2} \left(1 - \frac{\sqrt{2}}{2} \right) \\
 &= \frac{\pi R^4}{4} (2 - \sqrt{2})
 \end{aligned}$$

4. 换元 $x = \rho \sin \theta \cos \phi$
 $y = \rho \sin \theta \sin \phi$
 $z = \rho \cos \theta$
 $1 \leq \rho \leq 2 \quad 0 \leq \theta \leq \frac{\pi}{4} \quad 0 \leq \phi \leq 2\pi$

由对称性 $\Rightarrow \iiint_{\Omega} z y \, dx \, dy \, dz = 0$

$$\begin{aligned} \iiint_{\Omega} z(x^2 + y^2) \, dV &= \int_1^2 \int_0^{\frac{\pi}{4}} \int_0^{2\pi} \rho \cos \theta \cdot \rho^2 \sin^2 \theta \cdot \rho^2 \sin \theta \, d\phi \, d\theta \, d\rho \\ &= \int_1^2 \rho^5 \, d\rho \int_0^{\frac{\pi}{4}} \cos \theta \sin^3 \theta \, d\theta \int_0^{2\pi} d\phi \\ &= \frac{\rho^6}{6} \Big|_1^2 \cdot \frac{1}{4} \sin^4 \theta \Big|_0^{\frac{\pi}{4}} \cdot 2\pi \\ &= \frac{2^6 - 1}{6} \times \frac{1}{16} \times 2\pi \\ &= \frac{21}{16} \pi \end{aligned}$$

5. V 分布在第一、三卦限. (xy 同号, $z > 0$)

$$\begin{aligned} \iiint_V \frac{x^2 y^2}{z} \, dx \, dy \, dz &= 2 \iint_{\substack{c \leq xy \leq d \\ \alpha \leq \frac{y}{x} \leq \beta}} x^2 y^2 \left(\int_{\frac{x+y^2}{b}}^{\frac{x+y^2}{a}} \frac{1}{z} \, dz \right) dx \, dy \\ &= 2 \iint_{\substack{c \leq xy \leq d \\ \alpha \leq \frac{y}{x} \leq \beta}} x^2 y^2 \, dx \, dy \cdot \ln \frac{b}{a} \end{aligned}$$

换元 $u = xy$

$v = \frac{y}{x}$

$c \leq u \leq d \quad \alpha \leq v \leq \beta$

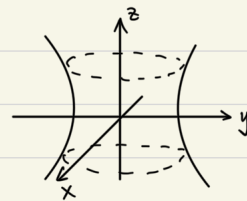
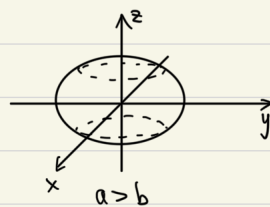
$\frac{\partial(u, v)}{\partial(x, y)} = y \cdot \frac{1}{x} - x \cdot \left(-\frac{y}{x^2}\right) = \frac{2y}{x}$

$\frac{\partial(x, y)}{\partial(u, v)} = \left(\frac{\partial(u, v)}{\partial(x, y)} \right)^{-1} = \frac{x}{2y} = \frac{1}{2v}$

$$\begin{aligned} \iint_{\substack{c \leq xy \leq d \\ \alpha \leq \frac{y}{x} \leq \beta}} x^2 y^2 \, dx \, dy &= \int_c^d \int_{\alpha}^{\beta} u^2 \cdot \frac{1}{2v} \, du \, dv \\ &= \frac{1}{2} \int_c^d u^2 \, du \int_{\alpha}^{\beta} \frac{1}{v} \, dv \\ &= \frac{1}{2} \cdot \frac{1}{3} (d^3 - c^3) \ln \frac{\beta}{\alpha} \end{aligned}$$

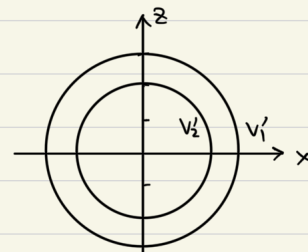
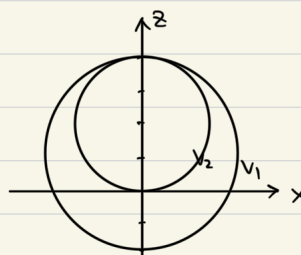
$$\begin{aligned} \iiint_V \frac{x^2 y^2}{z} \, dx \, dy \, dz &= 2 \cdot \frac{1}{2} \cdot \frac{1}{3} (d^3 - c^3) \ln \frac{\beta}{\alpha} \ln \frac{b}{a} \\ &= \frac{1}{3} (d^3 - c^3) \ln \frac{b}{a} \ln \frac{\beta}{\alpha} \end{aligned}$$

6. 由对称性 $\iiint_V x^2 y \, dx \, dy \, dz = 0$
 $\iiint_V xy z \, dx \, dy \, dz = 0$



$$\begin{aligned} \iiint_V z^2 \, dx \, dy \, dz &= \int_{-c}^c z^2 \left(\iint_{x^2 + y^2 \leq a^2 - \frac{b^2}{c^2} z^2} dx \, dy \right) dz \\ &= \int_{-c}^c \pi \left(a^2 - \frac{b^2}{c^2} z^2 \right) z^2 \, dz \\ &= 2\pi \left(a^2 \frac{c^3}{3} - \frac{a^2 b^2}{c^2} \frac{c^5}{5} \right) \\ &= \frac{2\pi}{15} c^3 (2a^2 + 3b^2) \end{aligned}$$

7.

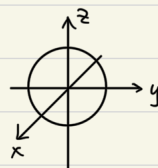


积分与 z 无关. 可以把两个球都移动至原点.

$$\begin{aligned} \iiint_V (x^2 + y^2) \, dV &= \left(\iiint_{V_1} - \iiint_{V_2} \right) (x^2 + y^2) \, dV \\ &= \left(\iiint_{V_1'} - \iiint_{V_2'} \right) (x^2 + y^2) \, dV \\ &= \iiint_{V'} (x^2 + y^2) \, dV \\ &= \int_2^3 \int_0^{\pi} \int_0^{2\pi} (\rho \sin \theta)^2 \cdot \rho^2 \sin \theta \, d\phi \, d\theta \, d\rho \\ &= \int_2^3 \rho^4 \, d\rho \int_0^{\pi} \sin^3 \theta \, d\theta \int_0^{2\pi} d\phi \\ &= \frac{\rho^5}{5} \Big|_2^3 \cdot 2 \int_0^{\pi} \sin^3 \theta \, d\theta \cdot 2\pi \\ &= 2\pi \cdot \frac{3^5 - 2^5}{5} \cdot 2 \cdot \frac{2!!}{3!!} \\ &= \frac{1688\pi}{15} \end{aligned}$$

8. 由对称性. $\iiint_V z e^{-(x^2 + y^2 + z^2)} \, dx \, dy \, dz$

$$\begin{aligned} \iiint_V (|x| + z) e^{-(x^2 + y^2 + z^2)} \, dx \, dy \, dz &= 2 \iiint_{V_1} x e^{-(x^2 + y^2 + z^2)} \, dx \, dy \, dz \\ &= 2 \int_1^2 \int_0^{\pi} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \rho \sin \theta \cos \phi e^{-\rho^2} \rho^2 \sin \theta \, d\phi \, d\theta \, d\rho \\ &= 2 \int_1^2 \rho^3 e^{-\rho^2} \, d\rho \int_0^{\pi} \sin^2 \theta \, d\theta \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos \phi \, d\phi \end{aligned}$$



$$\begin{aligned}
 &= 2 \cdot \frac{1}{2} \int_1^4 u e^{-u} du \cdot 2 \cdot \frac{1!!}{2!!} \cdot \frac{\pi}{2} \cdot 2 \quad + \times \quad e^{-x} \\
 &= 2 \cdot \frac{1}{2} (-u e^{-u} - e^{-u}) \Big|_1^4 \cdot \pi \quad - 1 \quad - e^{-x} \\
 &= \pi (2e^{-1} - 5e^{-4}) \quad 0 \quad + e^{-x}
 \end{aligned}$$

$$9. \iiint_V f(x, y, z) dx dy dz = \iiint_V (x^2 + y^2 + z^2) dx dy dz = I_1 + I_2 + I_3$$

$$\text{换元} \quad x = a r \sin \theta \cos \phi$$

$$y = b r \sin \theta \sin \phi$$

$$z = c r \cos \theta$$

$$0 \leq r \leq 1 \quad 0 \leq \theta \leq \frac{\pi}{2} \quad 0 \leq \phi \leq \frac{\pi}{2}$$

$$\frac{\partial(x, y, z)}{\partial(r, \theta, \phi)} = abc r^2 \sin \theta$$

$$\begin{aligned}
 I_1 &= \int_0^1 \int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{2}} a^2 r^2 \sin^2 \theta \cos^2 \phi \cdot abc r^2 \sin \theta dr d\theta d\phi \\
 &= abc \cdot a^2 \int_0^1 r^4 dr \int_0^{\frac{\pi}{2}} \sin^3 \theta d\theta \int_0^{\frac{\pi}{2}} \cos^2 \phi d\phi \\
 &= abc \cdot a^2 \cdot \frac{1}{5} \cdot \frac{2!!}{3!!} \cdot \frac{1!!}{2!!} \cdot \frac{\pi}{2} = \frac{\pi}{30} abc \cdot a^2
 \end{aligned}$$

$$\begin{aligned}
 I_2 &= \int_0^1 \int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{2}} b^2 r^2 \sin^2 \theta \sin^2 \phi \cdot abc r^2 \sin \theta dr d\theta d\phi \\
 &= abc \cdot b^2 \int_0^1 r^4 dr \int_0^{\frac{\pi}{2}} \sin^3 \theta d\theta \int_0^{\frac{\pi}{2}} \sin^2 \phi d\phi \\
 &= \frac{\pi}{30} abc \cdot b^2
 \end{aligned}$$

$$\begin{aligned}
 I_3 &= \int_0^1 \int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{2}} c^2 r^2 \cos^2 \theta \cdot abc r^2 \sin \theta dr d\theta d\phi \\
 &= abc \cdot c^2 \int_0^1 r^4 dr \int_0^{\frac{\pi}{2}} (\sin \theta - \sin^3 \theta) d\theta \int_0^{\frac{\pi}{2}} d\phi \\
 &= abc \cdot c^2 \cdot \frac{1}{5} \left(\frac{0!!}{1!!} - \frac{2!!}{3!!} \right) \cdot \frac{\pi}{2} \\
 &= \frac{\pi}{30} abc \cdot c^2
 \end{aligned}$$

$$\Rightarrow I = I_1 + I_2 + I_3 = \frac{\pi}{30} abc (a^2 + b^2 + c^2)$$

$$10. \text{换元} \quad x = \rho \sin \theta \cos \phi$$

$$y = \rho \sin \theta \sin \phi$$

$$z = \rho \cos \theta$$

$$\rho^4 \leq a^2 \rho^2 (\sin^2 \theta - \cos^2 \theta)$$

$$\rho \leq a \sqrt{\sin^2 \theta - \cos^2 \theta}$$

$$\sin^2 \theta \geq \cos^2 \theta \Rightarrow |\tan \theta| \geq 1 \Rightarrow \frac{\pi}{4} \leq \theta \leq \frac{3\pi}{4}$$

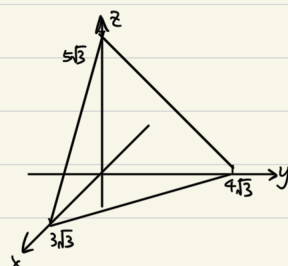
$$\begin{aligned}
 \iiint_V |z| dx dy dz &= \iiint_V \rho |\cos \theta| \rho^2 \sin \theta d\rho d\theta d\phi \\
 &= 2 \int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \int_0^{2\pi} \left[\int_0^{a \sqrt{\sin^2 \theta - \cos^2 \theta}} \rho^3 d\rho \right] \cos \theta \sin \theta d\theta d\phi \\
 &= 4\pi \int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \frac{a^4}{4} (\sin^2 \theta - \cos^2 \theta)^2 \cos \theta \sin \theta d\theta \\
 &= \pi a^4 \cdot \int_0^1 u^2 \cdot \frac{1}{4} du \\
 &= \frac{1}{4} \pi a^4 \cdot \frac{1}{3} \\
 &= \frac{\pi}{12} a^4
 \end{aligned}$$

$$11. \vec{n} \parallel \nabla F = \left(\frac{2x_0}{9}, \frac{2y_0}{16}, \frac{2z_0}{25} \right) = \left(\frac{2\sqrt{3}}{9}, \frac{\sqrt{3}}{6}, \frac{2\sqrt{3}}{15} \right)$$

$$\vec{n} = (20, 15, 12)$$

$$\pi: 20(x - \sqrt{3}) + 15(y - \frac{4\sqrt{3}}{3}) + 12(z - \frac{5\sqrt{3}}{3}) = 0$$

$$\mathbb{R}^3: 20x + 15y + 12z - 60\sqrt{3} = 0$$



$$15y + 12z \leq 60\sqrt{3}$$

$$y \leq 4(\sqrt{3} - \frac{y}{5})$$

$$\begin{aligned}
 \iiint_V x \left(\sqrt{3} - \frac{y}{4} - \frac{z}{5} \right) dx dy dz &= \iint_{D_{yz}} \left(\sqrt{3} - \frac{y}{4} - \frac{z}{5} \right) \int_0^{3(\sqrt{3} - \frac{y}{4} - \frac{z}{5})} x dx dy dz \\
 &= \frac{9}{2} \iint_{D_{yz}} \left(\sqrt{3} - \frac{y}{4} - \frac{z}{5} \right)^3 dy dz \\
 &= \frac{9}{2} \int_0^{5\sqrt{3}} \int_0^{4(\sqrt{3} - \frac{z}{5})} \left(\sqrt{3} - \frac{y}{4} - \frac{z}{5} \right)^3 dy dz \\
 &= \frac{9}{2} \int_0^{5\sqrt{3}} \int_0^{4(\sqrt{3} - \frac{z}{5})} \left[\left(\sqrt{3} - \frac{z}{5} \right)^3 - 3 \left(\sqrt{3} - \frac{z}{5} \right)^2 \cdot \frac{y}{4} \right. \\
 &\quad \left. + 3 \left(\sqrt{3} - \frac{z}{5} \right) \cdot \left(\frac{y}{4} \right)^2 - \left(\frac{y}{4} \right)^3 \right] dy dz \\
 &= \frac{9}{2} \int_0^{5\sqrt{3}} \left(4 - \frac{3}{4} \cdot \frac{16}{2} + 3 \cdot \frac{64}{16 \cdot 3} - \frac{256}{64 \cdot 4} \right) \left(\sqrt{3} - \frac{z}{5} \right)^4 dz \\
 &= \frac{9}{2} \int_0^{5\sqrt{3}} \left(\sqrt{3} - \frac{z}{5} \right)^4 dz \\
 &= \frac{9}{2} \cdot 5 \int_0^{\sqrt{3}} u^4 du \\
 &= \frac{9}{2} \cdot 5 \cdot \frac{(\sqrt{3})^5}{5} \\
 &= \frac{81}{2} \sqrt{3}
 \end{aligned}$$

$$12. (b). f(x, y, z) = x^2 + 2y - 2z + 3$$

在 $x^2 + y^2 + z^2 \leq 1$ (有界闭集) 上有最大、小值

$$\text{内部: } \begin{cases} f'_x = 2x = 0 \\ f'_y = 2 \neq 0 \\ f'_z = -2 \neq 0 \end{cases}$$

在内部无极值点.

$$F(x, y, z) = x^2 + 2y - 2z + 3 + \lambda(x^2 + y^2 + z^2 - 1)$$

$$\begin{cases} F'_x = 2x + 2\lambda x = 0 \\ F'_y = 2 + 2\lambda y = 0 \\ F'_z = -2 + 2\lambda z = 0 \\ x^2 + y^2 + z^2 = 1 \end{cases}$$

$$\text{若 } x=0, \lambda=1 \Rightarrow \lambda = \pm \sqrt{2}$$

$$(0, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}) \quad f = 3 + 2\sqrt{2}$$

$$(0, \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}) \quad f = 3 - 2\sqrt{2}$$

若 $x \neq 0$, 则 $\lambda = -1$.

$$\Rightarrow y=1, z=-1 \quad x^2 = -1 < 0, \text{ 矛盾!}$$

$$\text{因此, } f_{\max} = 3+2\sqrt{2} \quad f_{\min} = 3-2\sqrt{2}$$

$$\Rightarrow \sqrt{3-2\sqrt{2}} \cdot \frac{4}{3}\pi \leq \iiint_{\Omega} \sqrt{x^2+y-2z+3} \, dx dy dz \leq \sqrt{3+2\sqrt{2}} \cdot \frac{4}{3}\pi$$

$$\quad \quad \quad \parallel \quad \quad \quad \parallel$$

$$(\sqrt{2}-1) \frac{4}{3}\pi \quad \quad \quad (\sqrt{2}+1) \frac{4}{3}\pi$$

13. 不妨取 $S_0: -1 \leq x, y, z \leq 1$.

设动点 $P(x, y, z)$

$$S_P: -1+x \leq X \leq 1+x$$

$$-1+y \leq Y \leq 1+y$$

$$-1+z \leq Z \leq 1+z$$

$$x \geq 0 \text{ 时, } -1+x \leq X \leq 1$$

$$2-|x|$$

$$x < 0 \text{ 时 } -1 \leq X \leq 1+x$$

因此, $S_0 \cap S_P$ 的体积为 $(2-|x|)(2-|y|)(2-|z|)$

$$\Omega_1 = \{(x, y, z) \mid (2-x)(2-y)(2-z) \geq 4, 0 \leq x, y, z \leq 2\}$$

$$\begin{cases} u=2-x \\ v=2-y \\ w=2-z \end{cases}$$

$$\iiint_{\Omega_1} dV = \iiint_{\substack{0 < u, v, w \leq 2 \\ uvw \geq 4}} dV = \int_1^2 \left[\iint_{\substack{0 < v, w \leq 2 \\ vw \geq \frac{4}{u}}} dv dw \right] du = \int_1^2 \left(\int_{\frac{4}{u}}^2 \left(\int_{\frac{4}{uv}}^2 dw \right) dv \right) du$$

$$= \int_1^2 \left(\int_{\frac{4}{u}}^2 \left(2 - \frac{4}{uv} \right) dv \right) du$$

$$= \int_1^2 \left(4 \left(1 - \frac{1}{u} \right) - \frac{4}{u} \ln u \right) du$$

$$= 4 - 4 \ln 2 - 2 \ln^2 2$$

$$V(\Omega) = 8 \iiint_{\Omega_1} dV = 32 - 32 \ln 2 - 16 \ln^2 2$$

二、第一型曲线积分

$$\begin{array}{c} L_1 \\ \diagdown \quad \diagup \\ L_2 \quad L_1 \\ \diagup \quad \diagdown \\ L_1 \end{array} \quad x+y=1$$

$$1. \int_C (2x+y)^5 ds = \int_0^1 (2x)^5 dx + \int_0^1 (x+1)^5 \cdot \sqrt{2} dx + \int_0^1 y^5 dy$$

$$= 2^5 \cdot \frac{1}{6} + \sqrt{2} \cdot \frac{2^6-1}{6} + \frac{1}{6}$$

$$= \frac{33+63\sqrt{2}}{6}$$

$$2. \int_C (x^2 + x \cos x) ds = \int_C x^2 ds + \int_C x \cos x ds$$

$$= \int_C x^2 ds = \int_0^{2\pi} \cos^2 \theta \cdot d\theta$$

$$= 4 \cdot \frac{1!!}{2!!} \cdot \frac{\pi}{2}$$

$$= \pi$$

3. 方法一:

$$\begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = P \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

其中 P 为正交阵. 正交变换

$$u = \frac{x+y+z}{\sqrt{3}} = 0.$$

$$u^2 + v^2 + w^2 = x^2 + y^2 + z^2 = a^2$$

$$w = \frac{1}{\sqrt{6}}(x+y) + \frac{2}{\sqrt{6}}z$$

$$z = x+y = \sqrt{\frac{2}{3}} w$$

$$\Rightarrow \int_L z^2 ds = \frac{2}{3} \int_L w^2 ds = \frac{2}{3} \cdot \frac{1}{2} \int_L (w^2 + v^2) ds$$

$$= \frac{a^2}{3} \int_L ds$$

$$= \frac{2\pi}{3} a^3$$

方法二:

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} 1 & & \\ & 1 & \\ & & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

正交变换

$$x+y-z = x'+y'+z' = 0$$

$$x^2 + y^2 + z^2 = x'^2 + y'^2 + z'^2 = a^2$$

由对称性.

$$I = \frac{1}{3} \int_L (x'^2 + y'^2 + z'^2) ds$$

$$= \frac{1}{3} a^2 \cdot 2\pi a$$

$$= \frac{2\pi}{3} a^3$$

方法三:

平面 $x+y-z=0$ 法向量 $\vec{n} = (1, 1, -1)$

与三个坐标轴的夹角余弦值均为 $\frac{1}{\sqrt{3}}$.

即, 对于三个坐标轴完全对称.

$$\int_L x^2 ds = \int_L y^2 ds = \int_L z^2 ds$$

$$\Rightarrow \int_L z^2 ds = \frac{1}{3} \int_L (x^2 + y^2 + z^2) ds = \frac{1}{3} a^2 \cdot 2\pi a = \frac{2\pi}{3} a^3$$

补充: 求 $\int_L x^2 ds, \int_L y^2 ds, \int_L z^2 ds$, 其中 L 为圆周 $\begin{cases} x^2 + y^2 + z^2 = a^2 \\ Ax + By + Cz = 0 \end{cases}$

$$\vec{n} = \frac{1}{\sqrt{A^2+B^2+C^2}} (A, B, C)$$

$$\vec{e}_1 = \frac{1}{\sqrt{A^2+B^2}} (B, -A, 0)$$

$$\vec{e}_2 = \frac{1}{\sqrt{A^2+B^2}\sqrt{A^2+B^2+C^2}} (AC, BC, -(A^2+B^2))$$

$$\begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} \frac{A}{\sqrt{A^2+B^2+C^2}} & \frac{B}{\sqrt{A^2+B^2+C^2}} & \frac{C}{\sqrt{A^2+B^2+C^2}} \\ \frac{B}{\sqrt{A^2+B^2+C^2}} & \frac{-A}{\sqrt{A^2+B^2+C^2}} & 0 \\ \frac{AC}{\sqrt{A^2+B^2+C^2}} & \frac{BC}{\sqrt{A^2+B^2+C^2}} & \frac{-\sqrt{A^2+B^2}}{\sqrt{A^2+B^2+C^2}} \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

圆上任一点, $\vec{r}(\theta) = a \cos \theta \vec{e}_1 + a \sin \theta \vec{e}_2$

$$x(\theta) = \vec{r} \cdot \hat{x} = a \cos \theta \frac{B}{\sqrt{A^2+B^2}} + a \sin \theta \frac{AC}{\sqrt{A^2+B^2+C^2}}$$

$$y(\theta) = \vec{r} \cdot \hat{y} = a \cos \theta \frac{-A}{\sqrt{A^2+B^2}} + a \sin \theta \frac{BC}{\sqrt{A^2+B^2+C^2}}$$

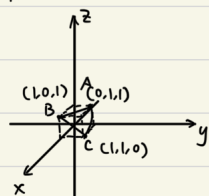
$$z(\theta) = \vec{r} \cdot \hat{z} = a \sin \theta \frac{-\sqrt{A^2+B^2}}{\sqrt{A^2+B^2+C^2}}$$

$$\int x^2 ds = \pi a^3 \cdot \frac{B^2+C^2}{A^2+B^2+C^2}$$

$$\int y^2 ds = \pi a^3 \cdot \frac{C^2+A^2}{A^2+B^2+C^2}$$

$$\int z^2 ds = \pi a^3 \cdot \frac{A^2+B^2}{A^2+B^2+C^2}$$

4.



$$L_1: x=t \quad y=1-t \quad z=1$$

$$\int_{L_1} xy ds = \int_0^1 t(1-t) \sqrt{2} dt = \sqrt{2} \cdot \left(\frac{1}{2} - \frac{1}{3}\right) = \frac{\sqrt{2}}{6}$$

$$L_2: x=1 \quad y=t \quad z=1-t$$

$$\int_{L_2} xy ds = \int_0^1 t \cdot \sqrt{2} dt = \sqrt{2} \cdot \frac{1}{2} = \frac{\sqrt{2}}{2}$$

$$L_3: y=1 \quad x=1-\sin \theta \quad z=1-\cos \theta$$

$$\int_{L_3} xy ds = \int_0^{\frac{\pi}{2}} (1-\sin \theta) d\theta = \frac{\pi}{2} - 1$$

$$\Rightarrow \int_L xy ds = \frac{\sqrt{2}}{6} + \frac{\sqrt{2}}{2} + \frac{\pi}{2} - 1 = \frac{2}{3}\sqrt{2} + \frac{\pi}{2} - 1$$

$$5. \quad \int_{L_{AB}} (x+y+z) ds = \int_0^2 (2+t) dt = 8 - \frac{2^2}{2} = 6$$

$$\int_{L_{BC}} (x+y+z) ds = \int_0^{2\pi} (2 \cos t + 2 \sin t + 1) \sqrt{5} dt$$

$$= \sqrt{5} \cdot \frac{1}{2} (2\pi)^2$$

$$= 2\sqrt{5} \pi^2$$

$$\Rightarrow \int_L (x+y+z) ds = 6 + 2\sqrt{5} \pi^2$$

$$6. \quad \int_L e^{\sqrt{x^2+y^2}} ds = e^2 \cdot \frac{1}{8} \cdot 2\pi \cdot 2 + \int_0^{\sqrt{2}} e^{\sqrt{2}t} \cdot \sqrt{2} dt + \int_0^2 e^t dt$$

$$= \frac{\pi}{2} e^2 + e^2 - 1 + e^2 - 1$$

$$= \left(\frac{\pi}{2} + 2\right) e^2 - 2$$



$$7. \quad L = L \cap \{(x,y) | x \geq 0, y \geq 0\}$$

$$x^2 + 3y = 1 \quad ds = \sqrt{1 + (-x^2)^2} dx$$

$$\int_L |x|^3 ds = 4 \int_0^1 x^3 \sqrt{1+x^4} dx = \int_0^1 \sqrt{1+u} du = \frac{2}{3} (1+u)^{\frac{3}{2}} \Big|_0^1 = \frac{2}{3} (2\sqrt{2} - 1)$$

三. 第一型曲面积分.

$$1. \quad z = \sqrt{1 - \frac{1}{2}x^2 - \frac{1}{2}y^2} \quad z z'_x = -x \quad z z'_y = -y$$

$$dS = \sqrt{1 + \left(\frac{-x}{z}\right)^2 + \left(\frac{-y}{z}\right)^2} dx dy$$

$$= \frac{1}{z} \sqrt{1 - \frac{1}{2}x^2 - \frac{1}{2}y^2} dx dy$$

$$D: x^2 + y^2 \leq 2$$

$$\iint_S z \sqrt{x^2+y^2+4z^2} dS = \iint_D z \sqrt{4-x^2-y^2} \cdot \frac{1}{2z} \sqrt{4-x^2-y^2} dx dy$$

$$= \iint_D \frac{1}{2} (4-x^2-y^2) dx dy$$

$$= \int_0^{\sqrt{2}} \int_0^{2\pi} \frac{1}{2} (4-r^2) r dr d\phi$$

$$= \int_0^2 \frac{1}{4} (4-u) du \int_0^{2\pi} d\phi$$

$$= \frac{\pi}{2} \cdot \frac{1}{2} u^2 \Big|_0^4$$

$$= 3\pi$$

$$2. \quad z = \sqrt{x^2+y^2}$$

$$dS = \sqrt{1 + \left(\frac{x}{\sqrt{x^2+y^2}}\right)^2 + \left(\frac{y}{\sqrt{x^2+y^2}}\right)^2} dx dy = \sqrt{2} dx dy$$

$$D: x^2 + y^2 \leq a^2$$

$$\iint_S \sqrt{x^2+y^2} dS = \iint_D \sqrt{x^2+y^2} \sqrt{2} dx dy$$

$$= \sqrt{2} \int_0^a r \cdot r dr \int_0^{2\pi} d\phi$$

$$= \sqrt{2} \cdot \frac{1}{3} a^3 \cdot 2\pi$$

$$= \frac{2\sqrt{2}}{3} \pi a^3$$

$$3. \quad x^2 + y^2 = 2 - \sqrt{x^2+y^2} \Rightarrow (\sqrt{x^2+y^2} + 2)(\sqrt{x^2+y^2} - 1) = 0$$

$$\Rightarrow x^2 + y^2 = 1$$

$$V = \iint_{x^2+y^2 \leq 1} \left(\int_{x^2+y^2}^{2-\sqrt{x^2+y^2}} dz \right) dx dy$$

$$= \iint_{x^2+y^2 \leq 1} (2 - \sqrt{x^2+y^2} - (x^2+y^2)) dx dy$$

$$= 2\pi \int_0^1 (2-r-r^2) r dr$$

$$= 2\pi \left(r^2 - \frac{r^3}{3} - \frac{r^4}{4} \right) \Big|_0^1$$

$$= 2\pi \left(1 - \frac{1}{3} - \frac{1}{4} \right)$$

$$= \frac{5\pi}{6}$$

$$dS_1 = \sqrt{1 + (2x)^2 + (2y)^2} dx dy \quad D_1: x^2 + y^2 \leq 1$$

$$S_1 = \iint_{x^2+y^2 \leq 1} \sqrt{1+4x^2+4y^2} dx dy = 2\pi \int_0^1 \sqrt{1+4r^2} \cdot r dr$$

$$= \frac{\pi}{4} \int_1^5 \sqrt{u} du$$

$$= \frac{\pi}{4} \cdot \frac{2}{3} u^{\frac{3}{2}} \Big|_1^5$$

$$= \frac{\pi}{6} (5\sqrt{5} - 1)$$

$$dS_2 = \sqrt{1 + \left(\frac{x}{\sqrt{x^2+y^2}}\right)^2 + \left(\frac{y}{\sqrt{x^2+y^2}}\right)^2} dx dy = \sqrt{2} dx dy \quad D_2: x^2+y^2 \leq 1$$

$$S_2 = \iint_{x^2+y^2 \leq 1} \sqrt{2} dx dy = \sqrt{2} \pi$$

$$S = S_1 + S_2 = \left[\frac{1}{6} (5\sqrt{5} - 1) + \sqrt{2} \right] \pi$$

$$4. \quad z = 1 - x - y$$

$$dS = \sqrt{1 + (-1)^2 + (-1)^2} dx dy = \sqrt{3} dx dy$$

$$D: x+y \leq 1 \quad x \geq 0, y \geq 0.$$

$$S = \iint_D x^2 y^3 (1-x-y) \sqrt{3} dx dy$$

$$= \int_0^1 \sqrt{3} x^2 \left(\int_0^{1-x} y^3 (1-x-y) dy \right) dx$$

$$= \int_0^1 \sqrt{3} x^2 \left((1-x) \cdot \frac{1}{4} (1-x)^4 - \frac{1}{5} (1-x)^5 \right) dx$$

$$= \frac{\sqrt{3}}{20} \int_0^1 x^2 (1-x)^5 dx$$

$$= \frac{\sqrt{3}}{20} B(3, 6)$$

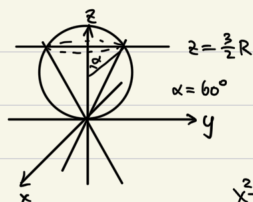
$$= \frac{\sqrt{3}}{20} \frac{\Gamma(3) \Gamma(6)}{\Gamma(9)}$$

$$= \frac{\sqrt{3}}{20} \frac{2! \times 5!}{8!}$$

$$= \frac{\sqrt{3}}{3360}$$

$$5. \quad x^2 + y^2 + (z-R)^2 = R^2$$

$$z^2 = 3x^2 + 3y^2$$



$$\Rightarrow \frac{z^2}{3} + (z-R)^2 = R^2$$

$$\Rightarrow \frac{4}{3} z^2 - 2Rz = 0$$

$$\Rightarrow \frac{4}{3} z \left(z - \frac{3}{2} R \right) = 0$$

$$x^2 + y^2 = \frac{3}{4} R^2 = \left(\frac{\sqrt{3}}{2} R \right)^2$$

$$S = 2\pi R^2 (1 - \cos \alpha) = 2\pi R^2 \left(1 - \frac{1}{2} \right) = \pi R^2$$

6.

$$\vec{r}'_\theta = \begin{bmatrix} r'(\theta) \sin \theta \cos \varphi + r(\theta) \cos \theta \cos \varphi \\ r'(\theta) \sin \theta \sin \varphi + r(\theta) \cos \theta \sin \varphi \\ r'(\theta) \cos \theta - r(\theta) \sin \theta \end{bmatrix}$$

$$\vec{r}'_\varphi = \begin{bmatrix} -r(\theta) \sin \theta \sin \varphi \\ r(\theta) \sin \theta \cos \varphi \\ 0 \end{bmatrix}$$

$$E = \vec{r}'_\theta \cdot \vec{r}'_\theta = r'^2 + r^2$$

$$F = \vec{r}'_\varphi \cdot \vec{r}'_\varphi = r^2 \sin^2 \theta$$

$$G = \vec{r}'_\theta \cdot \vec{r}'_\varphi = 0$$

$$|\vec{r}'_\theta \times \vec{r}'_\varphi| = \sqrt{EF - G^2} = \sqrt{r'^2 + r^2} \cdot r \sin \theta$$

$$dS = \sqrt{r'^2 + r^2} \cdot r \sin \theta d\theta d\varphi$$

$$\sigma(S) = \iint_D \sqrt{r'^2 + r^2} \cdot r \sin \theta d\theta d\varphi$$

$$\text{换元} \quad x = \rho \sin \theta \cos \phi$$

$$y = \rho \sin \theta \sin \phi$$

$$z = \rho \cos \theta$$

$$\rho^2 = \alpha^2 (\sin^2 \theta - \cos^2 \theta) \geq 0 \Rightarrow \tan^2 \theta \geq 1 \Rightarrow \frac{\pi}{4} \leq \theta \leq \frac{3\pi}{4}$$

$$S = \int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \int_0^{2\pi} \sqrt{\rho^2 + \rho^2} \rho \sin \theta d\theta d\phi$$

$$= 2\pi \int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \alpha^2 \sin \theta d\theta$$

$$= 2\sqrt{2} \pi \alpha^2$$

$$7. \quad z = \sqrt{x^2 + y^2}$$

$$dS = \sqrt{1 + \left(\frac{x}{\sqrt{x^2+y^2}}\right)^2 + \left(\frac{y}{\sqrt{x^2+y^2}}\right)^2} dx dy = \sqrt{2} dx dy$$

$$\iint_{\Sigma} |xyz| dS = 4 \iint_{\Sigma_1} xyz dS = 4 \iint_{D_1} xy \sqrt{x^2+y^2} \cdot \sqrt{2} dx dy$$

$$= 4 \iint_{x^2+y^2 \leq 1} r \cos \theta \cdot r \sin \theta \cdot r \sqrt{2} r dr d\theta$$

$$= 4\sqrt{2} \int_0^1 r^4 dr \int_0^{\frac{\pi}{2}} \sin \theta \cos \theta d\theta$$

$$= 4\sqrt{2} \cdot \frac{r^5}{5} \Big|_0^1 \cdot \frac{1}{2} \sin^2 \theta \Big|_0^{\frac{\pi}{2}}$$

$$= \frac{2\sqrt{2}}{5}$$

$$8. (1). \quad P(x_0, y_0, z_0)$$

$$\vec{n} \parallel \nabla F = (2x, 2y-z, 2z-y) \Big|_P \perp (0, 0, 1)$$

$$P: \begin{cases} 2z-y=0 \\ x^2+y^2+z^2-yz=1 \end{cases}$$

$$(2) \quad F'_x = 2x \quad F'_y = 2y-z \quad F'_z = 2z-y$$

$$dS = \sqrt{1 + \left(-\frac{F'_x}{F'_z}\right)^2 + \left(-\frac{F'_y}{F'_z}\right)^2} dx dy$$

$$= \frac{\sqrt{(2z-y)^2 + (2x)^2 + (2y-z)^2}}{|2z-y|} dx dy$$

$$= \frac{\sqrt{4x^2 + 5y^2 + 5z^2 - 8yz}}{|2z-y|} dx dy$$

$$= \frac{\sqrt{4 + y^2 + z^2 - 4yz}}{|2z-y|} dx dy$$

$$D: x^2 + y^2 + \left(\frac{y}{2}\right)^2 - y \cdot \frac{y}{2} \leq 1 \quad \&P \quad x^2 + \frac{3}{4} y^2 \leq 1$$

$$\iint_{\Sigma} \frac{(x+\sqrt{3}) |y-2z|}{\sqrt{4+y^2+z^2-4yz}} dS = \iint_D (x+\sqrt{3}) dx dy$$

$$= \sqrt{3} \iint_D dx dy$$

$$= \sqrt{3} \cdot \pi \cdot 1 \cdot \frac{2}{\sqrt{3}}$$

$$= 2\pi$$

四. 第二型曲线积分.

(-) Green 公式

$$\begin{aligned} 1. (1) I_1 &= \oint (xe^x + 3x^2y) dx + (x^3 + \sin y) dy \\ &= \iint_S (3x^2 - 3x^2) dx dy \\ &= 0 \end{aligned}$$

$$\begin{aligned} (2) (xe^x + 3x^2y) dx + (x^3 + \sin y) dy &= d((x-1)e^x + x^3y - \cos y) \\ I_2 &= (x-1)e^x + x^3y - \cos y \Big|_{(-1,0)}^{(2,3)} \\ &= (e^2 + 24 - \cos 3) - (-2e^{-1} - 1) \\ &= e^2 + 2e^{-1} + 25 - \cos 3 \end{aligned}$$

2. $\varepsilon > 0$ 充分小

$L_\varepsilon: 4x^2 + y^2 = \varepsilon^2$. 记 L 与 L_ε^- 围成的部分记为 S

$$\oint_{L+L_\varepsilon^-} \frac{-y dx + x dy}{4x^2 + y^2} = \iint_S \left(\frac{(4x^2+y^2) - x \cdot 8x}{(4x^2+y^2)^2} - \frac{(4x^2+y^2) + 2y}{(4x^2+y^2)^2} \right) dx dy = 0$$

$$\begin{aligned} I &= \oint_L \frac{-y dx + x dy}{4x^2 + y^2} = - \oint_{L_\varepsilon^-} \frac{-y dx + x dy}{4x^2 + y^2} \\ &= \oint_{L_\varepsilon} \frac{-y dx + x dy}{4x^2 + y^2} \\ &= \frac{1}{\varepsilon^2} \oint_{L_\varepsilon} -y dx + x dy \\ &= \frac{1}{\varepsilon^2} \iint_{4x^2+y^2 \leq \varepsilon^2} 2 dx dy \\ &= \frac{2}{\varepsilon^2} \cdot \pi \cdot \frac{\varepsilon}{2} \cdot \varepsilon \\ &= \pi \end{aligned}$$

3. $\varepsilon > 0$ 充分小

$L_\varepsilon: x^2 + y^2 = \varepsilon^2$. 记 L 与 L_ε^- 围成的部分记为 S

$$\oint_{L+L_\varepsilon^-} \frac{-y dx + x dy}{x^2 + y^2} = \iint_S \left(\frac{(x^2+y^2) - x \cdot 2x}{(x^2+y^2)^2} - \frac{-(x^2+y^2) + y \cdot 2y}{(x^2+y^2)^2} \right) dx dy = 0$$

$$\begin{aligned} \oint_L \frac{-y dx + x dy}{x^2 + y^2} &= - \oint_{L_\varepsilon^-} \frac{-y dx + x dy}{x^2 + y^2} \\ &= \oint_{L_\varepsilon} \frac{-y dx + x dy}{x^2 + y^2} \\ &= \frac{1}{\varepsilon^2} \oint_{L_\varepsilon} (-y dx + x dy) \\ &= \frac{2}{\varepsilon^2} \iint_{x^2+y^2 \leq \varepsilon^2} dx dy \\ &= \frac{2}{\varepsilon^2} \cdot \pi \varepsilon^2 \\ &= 2\pi \end{aligned}$$

$$I = \oint_L \frac{-y dx + x dy}{x^2 + y^2} = - \oint_{L_\varepsilon^-} \frac{-y dx + x dy}{x^2 + y^2} = -2\pi$$

4. $\varepsilon > 0$ 充分小

$L_\varepsilon: x^2 + 2y^2 = \varepsilon^2$ 记 L 与 L_ε^- 围成的部分记为 S

$$\oint_{L+L_\varepsilon^-} \frac{-y dx + x dy}{x^2 + 2y^2} = \iint_S \left(\frac{(x^2+2y^2) - x \cdot 2x}{(x^2+2y^2)^2} - \frac{-(x^2+2y^2) + y \cdot 4y}{(x^2+2y^2)^2} \right) dx dy = 0$$

$$\begin{aligned} I &= \oint_L \frac{-y dx + x dy}{x^2 + 2y^2} = - \oint_{L_\varepsilon^-} \frac{-y dx + x dy}{x^2 + 2y^2} \\ &= \oint_{L_\varepsilon} \frac{-y dx + x dy}{x^2 + 2y^2} \\ &= \frac{1}{\varepsilon^2} \oint_{L_\varepsilon} (-y dx + x dy) \\ &= \frac{1}{\varepsilon^2} \iint_{S_\varepsilon} 2 dx dy \\ &= \frac{1}{\varepsilon^2} \cdot 2 \cdot \pi \cdot \frac{\varepsilon}{\sqrt{2}} \cdot \varepsilon \\ &= \sqrt{2} \pi \end{aligned}$$

5. $\varepsilon > 0$ 充分小

$L_\varepsilon: Ax^2 + 2Bxy + Cy^2 = \varepsilon^2$

$\begin{bmatrix} A & B \\ B & C \end{bmatrix}$ 正定. $\Rightarrow L_\varepsilon$ 为椭圆

$\exists$ 可逆 P , s.t. $P^T A P = \begin{bmatrix} \lambda_1 & \\ & \lambda_2 \end{bmatrix}$

$$\lambda_1 u^2 + \lambda_2 v^2 = \varepsilon^2 \quad \lambda_1 \lambda_2 = |A| = AC - B^2$$

$$\Rightarrow \frac{u^2}{\frac{\varepsilon^2}{\lambda_1}} + \frac{v^2}{\frac{\varepsilon^2}{\lambda_2}} = 1$$

$$\Rightarrow S_\varepsilon = \pi \cdot \frac{\varepsilon}{\sqrt{\lambda_1}} \cdot \frac{\varepsilon}{\sqrt{\lambda_2}} = \frac{\pi \varepsilon^2}{\sqrt{AC - B^2}}$$

$$\oint_{L+L_\varepsilon^-} \frac{-y dx + x dy}{Ax^2 + 2Bxy + Cy^2} = \iint_S \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy = 0$$

$$\begin{aligned} \Rightarrow \oint_L \frac{-y dx + x dy}{Ax^2 + 2Bxy + Cy^2} &= - \oint_{L_\varepsilon^-} \frac{-y dx + x dy}{Ax^2 + 2Bxy + Cy^2} \\ &= \oint_{L_\varepsilon} \frac{-y dx + x dy}{Ax^2 + 2Bxy + Cy^2} \\ &= \frac{1}{\varepsilon^2} \oint_{L_\varepsilon} (-y dx + x dy) \\ &= \frac{1}{\varepsilon^2} \iint_{S_\varepsilon} 2 dx dy \\ &= \frac{2}{\varepsilon^2} S_\varepsilon \\ &= \frac{2}{\varepsilon^2} \frac{\pi \varepsilon^2}{\sqrt{AC - B^2}} \\ &= \frac{2\pi}{\sqrt{AC - B^2}} \end{aligned}$$

6. (1) 证: 对 \$t\$ 求偏导.

$$x f'_x(tx, ty) + y f'_y(tx, ty) = 2t f(x, y)$$

$$\text{令 } t \rightarrow 1 \Rightarrow x f'_x(x, y) + y f'_y(x, y) = 2 f(x, y)$$

$$\begin{aligned} (2) \int_L f(x, y) ds &= \frac{1}{2} \oint_L (x f'_x + y f'_y) ds \\ &= \oint_L \text{Pf} \cdot \left(\frac{x}{\sqrt{x^2+y^2}}, \frac{y}{\sqrt{x^2+y^2}} \right) ds = \left(\frac{x}{2}, \frac{y}{2} \right) \\ &\quad \vec{c} = \left(\frac{-y}{\sqrt{x^2+y^2}}, \frac{x}{\sqrt{x^2+y^2}} \right) = \left(-\frac{y}{2}, \frac{x}{2} \right) \\ &= \oint_L (-f'_y, +f'_x) \cdot \left(-\frac{y}{2}, \frac{x}{2} \right) ds \\ &= \oint_L (-f'_y, f'_x) \cdot \vec{c} ds \\ &= \oint_L -f'_y dx + f'_x dy \\ &= \iint_D \left(\frac{\partial}{\partial x} f'_x - \frac{\partial}{\partial y} (-f'_y) \right) dx dy \\ &= \iint_D \left(\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} \right) dx dy \end{aligned}$$

$$7. \oint_L -\frac{y}{f(x)} dx + x f(y) dy = \iint_S \left(f(y) + \frac{1}{f(x)} \right) dx dy$$

由对称性.

$$\iint_D f(y) dx dy = \iint_D f(x) dx dy$$

\$f(x)\$ 是正值连续函数.

$$\begin{aligned} \Rightarrow \oint_L -\frac{y}{f(x)} dx + x f(y) dy &= \iint_S \left(f(x) + \frac{1}{f(x)} \right) dx dy \\ &\geq 2 \iint_S dx dy = 2\pi \end{aligned}$$

$$8. \text{证: } \oint_L f(x, y) dx + g(x, y) dy = \iint_S \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) dx dy = 0$$

$$x = \cos \theta, y = \sin \theta$$

$$\Rightarrow 0 = \oint_L f(x, y) dx + g(x, y) dy = \int_0^{2\pi} [f(\cos \theta, \sin \theta)(-\sin \theta) + g(\cos \theta, \sin \theta) \cos \theta] d\theta$$

由于 \$f(x, y), g(x, y)\$ 在单位圆盘上有一阶连续偏导数. 因此, \$f(\cos \theta, \sin \theta)(-\sin \theta) + g(\cos \theta, \sin \theta) \cos \theta\$ 关于 \$\theta\$ 在 \$[0, 2\pi]\$ 上连续.

$$\Rightarrow \exists \theta_0 \in [0, 2\pi], \text{ s.t. } f(\cos \theta_0, \sin \theta_0) \sin \theta_0 = g(\cos \theta_0, \sin \theta_0) \cos \theta_0$$

即在单位圆盘上存在一点 \$(\xi, \eta)\$, s.t. \$f(\xi, \eta)\eta = g(\xi, \eta)\xi\$.

$$9. (1) \text{证: } \vec{c} = \begin{pmatrix} \cos \alpha \\ \sin \alpha \end{pmatrix}$$

$$\begin{aligned} \oint_L \frac{\partial u}{\partial n} ds &= \oint_L \nabla u \cdot \vec{n} ds = \oint_L \left(\frac{\partial f}{\partial x} \cos \alpha + \frac{\partial f}{\partial y} \sin \alpha \right) ds \\ &= \oint_L \left(-\frac{\partial f}{\partial y}, \frac{\partial f}{\partial x} \right) \cdot (-\sin \alpha, \cos \alpha) ds \\ &= \oint_L \left(-\frac{\partial f}{\partial y}, \frac{\partial f}{\partial x} \right) \cdot \vec{c} ds \end{aligned}$$

$$\begin{aligned} &= \oint_L -\frac{\partial f}{\partial y} dx + \frac{\partial f}{\partial x} dy \\ &= \iint_S \left(\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} \right) dx dy \\ &= 0 \end{aligned}$$

$$(2) \text{证: 由 (1)} \Rightarrow \oint_{\partial D} u \frac{\partial u}{\partial n} ds = A \oint_{\partial D} \frac{\partial u}{\partial n} ds = 0$$

$$\begin{aligned} \oint_{\partial D} u \frac{\partial u}{\partial n} ds &= \oint_{\partial D} \nabla u \cdot \vec{n} ds = \oint_{\partial D} \nabla u \cdot (\cos \alpha, \sin \alpha) ds \\ &= \oint_{\partial D} \left(u \frac{\partial u}{\partial x} \cos \alpha + u \frac{\partial u}{\partial y} \sin \alpha \right) ds \\ &= \oint_{\partial D} \left(-u \frac{\partial u}{\partial y}, u \frac{\partial u}{\partial x} \right) \cdot (-\sin \alpha, \cos \alpha) ds \\ &= \oint_{\partial D} \left(-u \frac{\partial u}{\partial y}, u \frac{\partial u}{\partial x} \right) \cdot \vec{c} ds \\ &= \oint_{\partial D} -u \frac{\partial u}{\partial y} dx + u \frac{\partial u}{\partial x} dy \\ &= \iint_D \left(\frac{\partial}{\partial x} \left(u \frac{\partial u}{\partial x} \right) + \frac{\partial}{\partial y} \left(u \frac{\partial u}{\partial y} \right) \right) dx dy \\ &= \iint_D \left[u \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) + \left(\left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial y} \right)^2 \right) \right] dx dy \\ 0 &= 0 + \iint_D \left(\left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial y} \right)^2 \right) dx dy \end{aligned}$$

$$\Rightarrow \frac{\partial u}{\partial x} = \frac{\partial u}{\partial y} = 0$$

$$\Rightarrow u(x, y) \equiv A, (x, y) \in D.$$

10. 证: 对 \$D\$ 内任意一个圆周 \$L\$

$$\begin{aligned} 0 &= \oint_L \vec{v} \cdot \vec{n} ds = \oint_L (P, Q) \cdot (\cos \alpha, \sin \alpha) ds \\ &= \oint_L (-Q, P) \cdot (-\sin \alpha, \cos \alpha) ds \\ &= \oint_L (-Q, P) \cdot \vec{c} ds \\ &= \oint_L -Q dx + P dy \\ &= \iint_S \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} \right) dx dy \end{aligned}$$

(反证法) 假设 \$\exists M_0 \in D\$, s.t. \$\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} > 0\$ (或 \$< 0\$)

由 \$\frac{\partial P}{\partial x}, \frac{\partial Q}{\partial y}\$ 的连续性以及连续函数的局部保号性

取 \$M_0\$ 的一个充分小的圆邻域 \$\Delta \subseteq D\$,

s.t. \$\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} > 0\$ (或 \$< 0\$) 在 \$\Delta\$ 上恒成立

$$\Rightarrow \iint_{\Delta} \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} \right) dx dy > 0 \text{ (或 } < 0 \text{)} \text{ 矛盾!}$$

故在 \$D\$ 内恒有 \$\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} = 0\$

$$\begin{aligned}
11. \quad \oint_{\partial D} \frac{\partial u}{\partial n} ds &= \oint_{\partial D} \nabla u \cdot \vec{n} ds = \oint_{\partial D} \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right) \cdot (\cos \alpha, \sin \alpha) ds \\
&= \oint_{\partial D} \left(-\frac{\partial f}{\partial y}, \frac{\partial f}{\partial x} \right) \cdot (-\sin \alpha, \cos \alpha) ds \\
&= \oint_{\partial D} \left(-\frac{\partial f}{\partial y}, \frac{\partial f}{\partial x} \right) \cdot \vec{e} ds \\
&= \oint_{\partial D} -\frac{\partial f}{\partial y} dx + \frac{\partial f}{\partial x} dy \\
&= \iint_D \left(\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} \right) dx dy \\
&= \iint_D e^{-(x^2+y^2)} dx dy \\
&= \int_0^1 \int_0^{2\pi} e^{-r^2} r dr d\theta \\
&= 2\pi \left[-\frac{1}{2} e^{-r^2} \right]_0^1 \\
&= \pi(1 - e^{-1})
\end{aligned}$$

$$\begin{aligned}
12. \quad (i) \quad x &= r \cos \theta, \quad y = r \sin \theta \\
x f'_x + y f'_y &= r \frac{\partial f}{\partial r}
\end{aligned}$$

$$\begin{aligned}
\iint_D (x f'_x + y f'_y) dx dy &= \int_0^1 \int_0^{2\pi} r \frac{\partial f}{\partial r} \cdot r dr d\theta \\
&= \int_0^{2\pi} \left(\int_0^1 r^2 \frac{\partial f}{\partial r} dr \right) d\theta
\end{aligned}$$

$$\begin{aligned}
\nabla^2 f &= e^{-(x^2+y^2)} = e^{-r^2} = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial f}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 f}{\partial \theta^2} \\
f(r, \theta) \text{ 关于 } \theta \text{ 周期性} \quad f(r, \theta) &= f(r, \theta + 2\pi)
\end{aligned}$$

两边对 θ 在 $[0, 2\pi]$ 上积分.

$$\int_0^{2\pi} \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial f}{\partial r} \right) d\theta + \int_0^{2\pi} \frac{1}{r^2} \frac{\partial^2 f}{\partial \theta^2} d\theta = \int_0^{2\pi} e^{-r^2} d\theta$$

$$\text{令 } J(r) = \int_0^{2\pi} f(r, \theta) d\theta$$

$$\Rightarrow \frac{1}{r} \frac{d}{dr} \left(r \frac{dJ}{dr} \right) = 2\pi e^{-r^2}$$

$$\Rightarrow \frac{d}{dr} \left(r \frac{dJ}{dr} \right) = 2\pi r e^{-r^2}$$

$$\Rightarrow r \frac{dJ}{dr} = \pi(1 - e^{-r^2})$$

$$\Rightarrow J(r) = \frac{\pi(1 - e^{-r^2})}{r}$$

$$\begin{aligned}
\int_0^{2\pi} \left(\int_0^1 r^2 \frac{\partial f}{\partial r} dr \right) d\theta &= \int_0^1 r^2 J'(r) dr = \int_0^1 r^2 \cdot \frac{\pi(1 - e^{-r^2})}{r} dr \\
&= \pi \int_0^1 (r - r e^{-r^2}) dr \\
&= \pi \left(\frac{1}{2} - \frac{1}{2}(1 - \frac{1}{e}) \right) \\
&= \frac{\pi}{2e}
\end{aligned}$$

$$13. \quad (i) \quad \forall (x_0, y_0) \in D, \quad \forall r > 0.$$

$$L: x = x_0 + r \cos \theta, \quad y = y_0 + r \sin \theta \quad (0 \leq \theta \leq \pi)$$

L : $(x_0 - r, y_0)$ 到 $(x_0 + r, y_0)$ 的直径线段.

$$\oint_{L+L} p(x, y) dx + q(x, y) dy = \iint_D \left(\frac{\partial q}{\partial x} - \frac{\partial p}{\partial y} \right) dx dy$$

$$\begin{aligned}
&\parallel \\
&\int_{x_0-r}^{x_0+r} p(x, y) dx + q(x, y) dy + \int_{x_0-r}^{x_0+r} p(x, y) dx \\
&\quad \parallel \\
&\int_{x_0-r}^{x_0+r} p(x, y) dx = \iint_D \left(\frac{\partial q}{\partial x} - \frac{\partial p}{\partial y} \right) dx dy
\end{aligned}$$

由积分中值定理. $\exists x^* \in [x_0 - r, x_0 + r]$, s.t.

$$\int_{x_0-r}^{x_0+r} p(x, y) dx = p(x^*, y_0) \cdot 2r$$

$$\text{令 } g(x, y) = \frac{\partial q}{\partial x} - \frac{\partial p}{\partial y}$$

由二重积分中值定理 $\exists (\xi, \eta) \in D$, s.t.

$$\iint_D g(x, y) dx dy = g(\xi, \eta) \cdot \frac{1}{2} \pi r^2$$

$$p(x^*, y_0) \cdot 2r = g(\xi, \eta) \cdot \frac{1}{2} \pi r^2$$

$$\Rightarrow p(x^*, y_0) = g(\xi, \eta) \cdot \frac{1}{4} \pi r$$

$$\text{令 } r \rightarrow 0^+, \text{ 则 } x^* \rightarrow x_0, (\xi, \eta) \rightarrow (x_0, y_0)$$

$$\Rightarrow p(x_0, y_0) = 0.$$

由 (x_0, y_0) 的任意性 $\Rightarrow \forall (x, y) \in \mathbb{R}^2, p(x, y) = 0$

$$\Rightarrow 0 = \iint_D \frac{\partial q}{\partial x} dx dy \stackrel{\text{二重积分中值定理}}{=} \frac{\partial q}{\partial x}(\xi', \eta') \cdot \frac{1}{2} \pi r^2$$

$$\Rightarrow \frac{\partial q}{\partial x}(\xi', \eta') = 0$$

$$\text{令 } r \rightarrow 0^+ \Rightarrow \frac{\partial q}{\partial x}(x_0, y_0) = 0$$

由 (x_0, y_0) 的任意性 $\Rightarrow \forall (x, y) \in \mathbb{R}^2, \frac{\partial q}{\partial x} = 0$.

$$14. \quad (1) \quad (i) \quad \oint_L x f(x, y) dy = \iint_S \left(f(x, y) + x \frac{\partial f}{\partial x} \right) dx dy$$

$$\Rightarrow \iint_D f(x, y) dx dy = \oint_L x f(x, y) dy - \iint_S x \frac{\partial f}{\partial x} dx dy$$

$$\begin{aligned}
(2) \quad (i) \quad \iint_D f^2 dx dy &= \oint_L x f^2 dy - \iint_S x \cdot 2f \frac{\partial f}{\partial x} dx dy \\
&= - \iint_S 2xf \frac{\partial f}{\partial x} dx dy
\end{aligned}$$

$$\text{同理, } \iint_D f^2 dx dy = - \iint_S 2yf \frac{\partial f}{\partial y} dx dy$$

$$\Rightarrow \iint_D f^2 dx dy = - \iint_S f \cdot (x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y}) dx dy$$

$$\Rightarrow (\iint_D f^2 dx dy)^2 = (\iint_S f \cdot (x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y}) dx dy)^2$$

$$\leq (\iint_S f^2 dx dy) (\iint_S (x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y})^2 dx dy)$$

$$\text{若 } \iint_D f^2 dx dy = 0 \quad \checkmark$$

$$\text{若 } \iint_D f^2 dx dy \neq 0, \text{ 则}$$

$$\iint_S f^2 dx dy \leq \iint_S (x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y})^2 dx dy$$

$$(x, y) \cdot (\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y})^2 \leq (x^2 + y^2) \left((\frac{\partial f}{\partial x})^2 + (\frac{\partial f}{\partial y})^2 \right)$$

$$\Rightarrow \iint_S f^2 dx dy \leq \iint_S (x^2 + y^2) \left((\frac{\partial f}{\partial x})^2 + (\frac{\partial f}{\partial y})^2 \right) dx dy$$

$$\leq \left(\max_{(x,y) \in D} \sqrt{x^2 + y^2} \right)^2 \iint_D \left((\frac{\partial f}{\partial x})^2 + (\frac{\partial f}{\partial y})^2 \right) dx dy$$

$$15. \quad I = \int_0^R \int_0^{2\pi} f(r \cos \theta, r \sin \theta) \cdot r dr d\theta$$

$$= \int_0^R r \left(\int_0^{2\pi} f(r \cos \theta, r \sin \theta) d\theta \right) dr$$

$$\text{令 } g(r) = \int_0^{2\pi} f(r \cos \theta, r \sin \theta) d\theta$$

$$h(r, \theta) = f(r \cos \theta, r \sin \theta)$$

在 $[0, R] \times [0, 2\pi]$ 上连续, 且 $\frac{\partial h}{\partial r}$ 连续

由含参变量积分的可微性:

$$g'(r) = \int_0^{2\pi} \frac{\partial f}{\partial r} d\theta = \int_0^{2\pi} (f'_x, f'_y) \cdot (\cos \theta, \sin \theta) d\theta$$

$$= \frac{1}{r} \oint_{L_r} (f'_y, f'_x) \cdot (-\sin \theta, \cos \theta) ds$$

$$= \frac{1}{r} \oint_{L_r} -f'_y dx + f'_x dy$$

$$= \frac{1}{r} \iint_{S_r} \left(\frac{\partial f}{\partial x^2} + \frac{\partial f}{\partial y^2} \right) dx dy$$

$$= \frac{1}{r} \iint_{S_r} (x^2 + y^2) dx dy$$

$$= \frac{1}{r} \iint_{x^2 + y^2 \leq r^2} (x^2 + y^2) dx dy$$

$$= \frac{1}{r} \int_0^r \int_0^{2\pi} \rho^2 d\rho d\theta$$

$$= \frac{1}{r} \cdot 2\pi \cdot \frac{r^4}{4}$$

$$= \frac{\pi}{2} r^3$$

$$g(0) = \int_0^{2\pi} f(0, 0) d\theta = 2\pi f(0, 0)$$

$$\Rightarrow g(r) = g(0) + \frac{\pi}{2} \int_0^r r^3 dr = 2\pi f(0, 0) + \frac{\pi}{8} r^4$$

$$I = \int_0^R r g(r) dr = \int_0^R r \left(2\pi f(0, 0) + \frac{\pi}{8} r^4 \right) dr$$

$$= \pi R^2 f(0, 0) + \frac{\pi}{78} R^6$$

(二) Stokes 公式

$$1. \quad \vec{n} = \frac{1}{\sqrt{3}} (1, 1, 1)$$

$$\int_L (y-z) dx + (z-x) dy + (x-y) dz$$

$$\stackrel{\text{Stokes}}{\underset{\text{公式}}{=}} \iint_S \frac{1}{\sqrt{3}} \begin{vmatrix} 1 & 1 & 1 \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y-z & z-x & x-y \end{vmatrix} dS$$

$$= -2\sqrt{3} \iint_S dS$$

$$= -2\sqrt{3} \pi a^2$$

$$2. \quad \vec{n} = \frac{1}{\sqrt{3}} (1, 1, 1)$$

$$(\nabla \times \vec{F}) \cdot \vec{n} = \frac{1}{\sqrt{3}} (-8x - 4y - 6z)$$

$$= \frac{1}{\sqrt{3}} (-2x + 2y - 1z)$$

$$dS = \sqrt{1+2^2+2^2} = \sqrt{3} dx dy$$

$$\oint_L (y^2-z^2) dx + (2z^2-x^2) dy + (3x^2-y^2) dz$$

$$\stackrel{\text{Stokes}}{\underset{\text{公式}}{=}} \iint_D \frac{1}{\sqrt{3}} (-2x + 2y - 1z) \sqrt{3} dx dy$$

$$= -12 \iint_D dx dy$$

$$= -24$$

$$D: |x| + |y| \leq 1$$

$$3. \quad \vec{n} \parallel \nabla F = (2x, 2y, 2z)$$

$$\vec{n} = (x, y, z)$$

$$\oint_L xz dy - yz dx \stackrel{\text{Stokes}}{\underset{\text{公式}}{=}} \iint_S \begin{vmatrix} x & y & z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ -yz & xz & 0 \end{vmatrix} dS$$

$$= \iint_S (2z^2 - z^2) dS$$

$$D: x^2 + y^2 \leq x$$

$$= \iint_D (2 - 3x^2 - 3y^2) \cdot \frac{1}{\sqrt{1-x^2-y^2}} dx dy$$

$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \int_0^{\cos \theta} \left(\frac{-1}{\sqrt{1-r^2}} + 3\sqrt{1-r^2} \right) r dr d\theta$$

$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\sqrt{1-u} - 2(1-u)^{\frac{3}{2}} \right) \Big|_0^{\cos^2 \theta} d\theta$$

$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (|\sin \theta| - |\sin \theta|^3) d\theta$$

$$= 2 \int_0^{\frac{\pi}{2}} (\sin \theta - \sin^3 \theta) d\theta$$

$$= 2 \times \left(\frac{0!!}{1!!} - \frac{2!!}{3!!} \right)$$

$$= \frac{2}{3}$$

$$4. \oint_L (y^2+z^2) dx + (z^2+x^2) dy + (x^2+y^2) dz$$

$$\vec{n} = -\left(\frac{x-z}{2}, \frac{y}{2}, \frac{z}{2}\right)$$

$$(\nabla \times \vec{F}) \cdot \vec{n} = - \begin{vmatrix} \frac{x-z}{2} & \frac{y}{2} & \frac{z}{2} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2+z^2 & z^2+x^2 & x^2+y^2 \end{vmatrix} = -2z$$

$$dS = \frac{z}{2} dx dy$$

$$\oint_L (y^2+z^2) dx + (z^2+x^2) dy + (x^2+y^2) dz$$

$$\frac{\text{Stokes}}{\text{1. 2. 3.}} \iint_S -2z dS$$

$$= \iint_D -2z \cdot \frac{z}{2} dx dy$$

$$= -4 \iint_D dx dy$$

$$= -4\pi$$

$$D: x^2+y^2 \leq 2x$$

$$5. \vec{n} = \frac{1}{\sqrt{2}}(1, 0, 1)$$

$$dS = \sqrt{2} dx dy$$

$$(\nabla \times \vec{F}) \cdot \vec{n} = \frac{1}{\sqrt{2}} \begin{vmatrix} 1 & 0 & 1 \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y & z & x \end{vmatrix} = -\sqrt{2} \quad 2x^2 + y^2 = 9$$

$$\oint_L y dx + z dy + x dz \stackrel{\text{Stokes}}{\text{1. 2. 3.}} -2 \iint_D dx dy$$

$$= -2\pi \cdot \frac{3}{\sqrt{2}} \cdot 3$$

$$= -9\sqrt{2}\pi$$

五. 第一型曲面积分.

(一). Gauss 公式.

$$1. \iint_{\Sigma+\Sigma_0} x^2 dy dz + (y^2+z^2+1) dx dy = \iiint_V (2x+1) dx dy dz$$

$$= 0 + \frac{2}{3}\pi$$

$$\Rightarrow \iint_{\Sigma} x^2 dy dz + (y^2+z^2+1) dx dy = \frac{2}{3}\pi - \iint_{\Sigma_0} x^2 dy dz + (y^2+z^2+1) dx dy$$

$$= \frac{2}{3}\pi - \iint_D (y^2+z^2+1) (-1) dx dy$$

$$= \frac{2}{3}\pi + \iint_D (y^2+z^2+1) dx dy$$

$$= \frac{2}{3}\pi + 0 + \pi$$

$$= \frac{5}{3}\pi$$

$$2. \iint_{S^++S_1} 2x^2 dy dz + 2y^2 dz dx + 3(z^2-1) dx dy$$

$$= \iiint_V (6x^2 + 6y^2 + 6z) dx dy dz$$

$$= 6 \iiint_V (x^2+y^2) dx dy dz + 6 \iiint_V z dx dy dz$$

$$\iiint_V (x^2+y^2) dx dy dz = \iint_{x^2+y^2 \leq 1} (x^2+y^2) \int_0^1 dz dx dy$$

$$= \iint_{x^2+y^2 \leq 1} (x^2+y^2) (1-\sqrt{x^2+y^2}) dx dy$$

$$= \int_0^1 \int_0^{2\pi} r^2(1-r) r dr d\theta$$

$$= \int_0^1 (r^3 - r^4) dr \int_0^{2\pi} d\theta$$

$$= \left(\frac{1}{4} - \frac{1}{5}\right) \cdot 2\pi$$

$$= \frac{\pi}{10}$$

$$\iiint_V z dx dy dz = \iint_{x^2+y^2 \leq 1} \left(\int_0^1 z dz\right) dx dy$$

$$= \iint_{x^2+y^2 \leq 1} \left(\frac{1}{2} - \frac{x^2+y^2}{2}\right) dx dy$$

$$= \int_0^1 \int_0^{2\pi} \frac{1}{2}(1-r^2) r dr d\theta$$

$$= \int_0^1 \frac{1}{2}(r-r^3) dr \int_0^{2\pi} d\theta$$

$$= \frac{1}{2} \left(\frac{1}{2} - \frac{1}{4}\right) \cdot 2\pi$$

$$= \frac{\pi}{4}$$

$$\iint_{S^+} 2x^2 dy dz + 2y^2 dz dx + 3(z^2-1) dx dy$$

$$= 6\pi \left(\frac{\pi}{10} + \frac{\pi}{4}\right) - \iint_{S_1} 3(z^2-1) dx dy$$

$$= \frac{21\pi}{10} - 0$$

$$= \frac{21}{10}\pi$$

$$3. \iint_{S^++S_0} (2x+z) dy dz + z dx dy$$

$$= \iiint_V (2+1) dx dy dz$$

$$= 3 \iiint_V dx dy dz$$

$$= 3 \iint_{x^2+y^2 \leq 1} \left(\int_0^1 dz\right) dx dy$$



$$= 3 \iint_{x^2+y^2 \leq 1} (1 - (x^2+y^2)) dx dy$$

$$= 3 \int_0^1 \int_0^{2\pi} (1-r^2) r dr d\theta$$

$$= 3 \int_0^1 (r-r^3) dr \int_0^{2\pi} d\theta$$

$$= 3 \cdot \left(\frac{1}{2} - \frac{1}{4}\right) \cdot 2\pi$$

$$= \frac{3\pi}{2}$$

$$\iint_S (2x+z) dy dz + z dx dy = - \iint_{S^-} (2x+z) dy dz + z dx dy$$

$$= \iint_{S_0} z dx dy - \frac{3\pi}{2}$$

$$= \iint_{x^2+y^2 \leq 1} dx dy - \frac{3\pi}{2}$$

$$= \pi - \frac{3\pi}{2}$$

$$= -\frac{\pi}{2}$$

$$4. \iint_S \frac{x^3 dy dz + y^3 dz dx + (z^3 + z^2) dx dy}{\sqrt{x^2+y^2+z^2}}$$

$$= \frac{1}{2} \iint_S x^3 dy dz + y^3 dz dx + (z^3 + z^2) dx dy$$

$$\iint_{S^+ \cup S_0} x^3 dy dz + y^3 dz dx + (z^3 + z^2) dx dy = \iiint_V (3x^2 + 3y^2 + 3z^2 + 2z) dx dy dz$$

$$= 3 \iiint_V (x^2 + y^2 + z^2) dx dy dz + 2 \iiint_V z dx dy dz$$

$$= 3 \int_0^2 \int_0^\pi \int_0^{2\pi} r^2 \cdot r^2 \sin \theta dr d\theta d\phi + 2 \iint_{x^2+y^2 \leq 4} \left(\int_0^{\sqrt{4-x^2-y^2}} z dz \right) dx dy$$

$$= 3 \cdot 4\pi \int_0^2 r^4 dr + \iint_{x^2+y^2 \leq 4} (4-x^2-y^2) dx dy$$

$$= 3 \cdot 4\pi \cdot \frac{16}{5} + 16\pi - \int_0^2 \int_0^{2\pi} r^2 \cdot r dr d\theta$$

$$= \frac{192\pi}{5} + 16\pi - 8\pi$$

$$= \frac{232\pi}{5}$$

$$\iint_S x^3 dy dz + y^3 dz dx + (z^3 + z^2) dx dy = - \iint_{S^-} x^3 dy dz + y^3 dz dx + (z^3 + z^2) dx dy$$

$$= -\frac{232\pi}{5} + \iint_{S_0} (z^3 + z^2) dx dy$$

$$= -\frac{232\pi}{5}$$

$$\Rightarrow \iint_S \frac{x^3 dy dz + y^3 dz dx + (z^3 + z^2) dx dy}{\sqrt{x^2+y^2+z^2}}$$

$$= \frac{1}{2} \iint_S x^3 dy dz + y^3 dz dx + (z^3 + z^2) dx dy$$

$$= -\frac{116\pi}{5}$$

$$5. S_E: x^2 + y^2 + z^2 = z^2 \quad z > 0 \text{ 球面}$$

$$\iint_{S^+ \cup S_E^-} \frac{x dy dz + y dz dx + z dx dy}{(x^2 + y^2 + z^2)^{\frac{3}{2}}} \stackrel{\text{Gauss}}{\underset{\text{d} \vec{z}}{=}} \iiint_V 0 \cdot dV = 0$$

$$\Rightarrow \iint_{S^+} \frac{x dy dz + y dz dx + z dx dy}{(x^2 + y^2 + z^2)^{\frac{3}{2}}} = \iint_{S_E} \frac{x dy dz + y dz dx + z dx dy}{(x^2 + y^2 + z^2)^{\frac{3}{2}}}$$

$$= \frac{1}{z^3} \iint_{S_E} x dy dz + y dz dx + z dx dy$$

$$\stackrel{\text{Gauss}}{\underset{\text{d} \vec{z}}{=}} \frac{1}{z^3} \iiint_{V_E} (1+1+1) dx dy dz$$

$$= \frac{1}{z^3} \cdot 3 \cdot \frac{4}{3} \pi z^3$$

$$= 4\pi$$

$$6. \iint_{S^+ \cup S_0} \vec{v} \cdot d\vec{S} = \iiint_V (0+0+0) dV = 0$$

$$\Rightarrow \iint_{S^+} \vec{v} \cdot d\vec{S} = - \iint_{S_0} \vec{v} \cdot d\vec{S}$$

$$= - \iint_{x^2+y^2 \leq 1} 2 \cdot (-1) dx dy$$

$$= 2 \iint_{x^2+y^2 \leq 1} dx dy$$

$$= 2\pi$$

$$7. \iint_{S^+ \cup S_0} (x+y) dy dz + (y+z) dz dx + (z+1) dx dy$$

$$\stackrel{\text{Gauss}}{\underset{\text{d} \vec{z}}{=}} \iiint_V (1+1+1) dx dy dz$$

$$= 3 \cdot \frac{2}{3} \pi R^3$$

$$= 2\pi R^3$$

$$\iint_{S^+} (x+y) dy dz + (y+z) dz dx + (z+1) dx dy$$

$$= 2\pi R^3 - \iint_{S_0} (z+1) dx dy$$

$$= 2\pi R^3 - \iint_{x^2+y^2 \leq R^2} (-1) dx dy$$

$$= 2\pi R^3 + \pi R^2$$

$$8. \iint_S (x-y+z) dy dz + (y-z+x) dz dx + (z-x+y) dx dy$$

$$\stackrel{\text{Gauss}}{\underset{\text{d} \vec{z}}{=}} \iiint_V (1+1+1) dx dy dz$$

$$= 3V$$

换元 $u = x - y + z$

$$v = y - z + x \quad \left| \frac{\partial(u, v, w)}{\partial(x, y, z)} \right| = 4$$

$$w = z - x + y \quad \left| \frac{\partial(x, y, z)}{\partial(u, v, w)} \right| = \frac{1}{4}$$

$$|u| + |v| + |w| = 1$$

$$V = \iiint_V dx dy dz = \iiint_{V'} \frac{1}{4} du dv dw$$

$$= 8 \iiint_{V'} \frac{1}{4} du dv dw$$

$$= 2 \int_0^1 \int_0^{1-w} \int_0^{1-v-w} du dv dw$$

$$= 2 \int_0^1 \int_0^{1-w} (1-v-w) dv dw$$

$$= 2 \int_0^1 \left((1-w)^2 - \frac{1}{2}(1-w)^2 \right) dw$$

$$= \int_0^1 (1-w)^2 dw$$

$$= \int_0^1 t^2 dt$$

$$= \frac{1}{3}$$

$$\oint_S (x-y+z) dy dz + (y-z+x) dz dx + (z-x+y) dx dy = 3V = 1$$

9. $0 < a < 1$ 时, 原点不在曲面 Σ 内.

$$I = \oint_{\Sigma} \frac{xdydz + ydzdx + zdx dy}{(x^2 + y^2 + 4z^2)^{\frac{3}{2}}} \stackrel{\text{Gauss}}{\underset{V}{=}} \iiint_V 0 dx dy dz = 0$$

$a > 1$ 时, $S_\varepsilon: x^2 + y^2 + 4z^2 = \varepsilon^2$. $\varepsilon > 0$ 充分小.

$$\oint_{\Sigma + S_\varepsilon^-} \frac{xdydz + ydzdx + zdx dy}{(x^2 + y^2 + 4z^2)^{\frac{3}{2}}} \stackrel{\text{Gauss}}{\underset{V}{=}} \iiint_V 0 dx dy dz = 0$$

$$\Rightarrow I = \oint_{\Sigma} \frac{xdydz + ydzdx + zdx dy}{(x^2 + y^2 + 4z^2)^{\frac{3}{2}}}$$

$$= \oint_{S_\varepsilon} \frac{xdydz + ydzdx + zdx dy}{(x^2 + y^2 + 4z^2)^{\frac{3}{2}}}$$

$$= \frac{1}{\varepsilon^3} \oint_{S_\varepsilon} x dy dz + y dz dx + z dx dy$$

$$= \frac{1}{\varepsilon^3} \iiint_{V_\varepsilon} (u+1) dx dy dz$$

$$= \frac{1}{\varepsilon^3} \cdot 3 \cdot \frac{4}{3} \pi \varepsilon \cdot \frac{\varepsilon}{2}$$

$$= 2\pi$$

$$10. \iint_{S+S_1} 2(1+x) dy dz + yz dx dy$$

$$\stackrel{\text{Gauss}}{\underset{V}{=}} \iiint_V (2+y) dx dy dz$$

$$= 2 \iiint_V dx dy dz + 0$$

$$= 2 \iint_{y^2+z^2 \leq 1} \left(\int_{y^2+z^2}^1 dx \right) dy dz$$

$$= 2 \iint_{y^2+z^2 \leq 1} (1 - (y^2+z^2)) dy dz$$

$$= 2 \int_0^1 \int_0^{2\pi} (1-r^2) r dr d\theta$$

$$= 4\pi \int_0^1 (r-r^3) dr$$

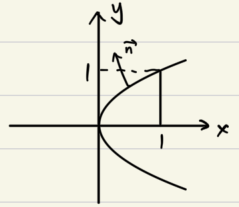
$$= \pi$$

$$\iint_S 2(1+x) dy dz + yz dx dy = \pi - \iint_{S_0} 2(1+x) dy dz$$

$$= \pi - \iint_{y^2+z^2 \leq 1} 4 dy dz$$

$$= \pi - 4\pi$$

$$= -3\pi$$



$$11. \iiint_S x^n dy dz + y^n dz dx + z^n dx dy$$

$$\stackrel{\text{Gauss}}{\underset{V}{=}} n \iiint_V (x^{n-1} + y^{n-1} + z^{n-1}) dx dy dz$$

$$n \text{ 偶} \quad I = 0$$

$$n \text{ 奇} \quad I = n \iiint_V (x^{n-1} + y^{n-1} + z^{n-1}) dx dy dz$$

$$V': u^2 + v^2 + w^2 \leq 1 \quad = n \iiint_{V'} ((au)^{n-1} + (bv)^{n-1} + (cw)^{n-1}) abc du dv dw$$

$$= \frac{1}{3} nabc (a^{n-1} + b^{n-1} + c^{n-1}) \iiint_{V'} (u^{n-1} + v^{n-1} + w^{n-1}) du dv dw$$

$$u = r \sin \theta \cos \phi$$

$$v = r \sin \theta \sin \phi$$

$$w = r \cos \theta$$

$$\iiint_{V'} (u^{n-1} + v^{n-1} + w^{n-1}) du dv dw$$

$$= \int_0^1 \int_0^\pi \int_0^{2\pi} r^{n-1} (\sin^{n-1} \theta \cos^{n-1} \phi + \sin^{n-1} \theta \sin^{n-1} \phi + \cos^{n-1} \theta) r^2 \sin \theta dr d\theta d\phi$$

$$= \int_0^1 r^{n+1} dr \left(\int_0^\pi \sin^n \theta d\theta \cdot \left(\int_0^{2\pi} \cos^{n-1} \phi d\phi + \int_0^{2\pi} \sin^{n-1} \phi d\phi \right) \right)$$

$$+ \int_0^\pi \cos^{n-1} \theta \sin \theta d\theta \cdot \int_0^{2\pi} d\phi$$

$$= \frac{1}{n+2} \left(2x \frac{(n-1)!!}{n!!} \times 2x(4x \frac{(n-2)!!}{(n-1)!!} \frac{\pi}{2} + \frac{2}{n} \cdot 2\pi) \right)$$

$$= \frac{12\pi}{n(n+2)}$$

$$\Rightarrow I = \frac{1}{3} nabc (a^{n-1} + b^{n-1} + c^{n-1}) \frac{12\pi}{n(n+2)}$$

$$= \frac{4\pi abc}{n+2} (a^{n-1} + b^{n-1} + c^{n-1})$$

$$I = \begin{cases} 0 & n \text{ 偶} \\ \frac{4\pi abc}{n+2} (a^{n-1} + b^{n-1} + c^{n-1}) & n \text{ 奇} \end{cases}$$

$$12. \iint_{S+S_0} P dy dz + R dx dy = \iiint_V \left(\frac{\partial P}{\partial x} + \frac{\partial R}{\partial z} \right) dx dy dz$$

||

$$\iint_S P dy dz + R dx dy + \iint_{S_0} R dx dy$$

||

$$0 + \iint_{x^2+y^2 \leq r^2} R(-1) dx dy$$

$$\Rightarrow - \iiint_{x^2+y^2 \leq r^2} R dx dy = \iiint_V \left(\frac{\partial P}{\partial x} + \frac{\partial R}{\partial z} \right) dx dy dz$$

由三重积分中值定理.

$$\exists (\xi_1, \eta_1) \in D, \text{ s.t. } \iint_D R(x, y, z_0) dx dy = R(\xi_1, \eta_1, z_0) \cdot \pi r^2$$

由三重积分中值定理

$$\exists (\xi_2, \eta_2, \zeta_2) \in V, \text{ s.t.}$$

$$\iiint_V \left(\frac{\partial P}{\partial x} + \frac{\partial R}{\partial z} \right) dx dy dz = \left(\frac{\partial P}{\partial x} + \frac{\partial R}{\partial z} \right) \Big|_{(\xi_2, \eta_2, \zeta_2)} \cdot \frac{2}{3} \pi r^2$$

$$\Rightarrow -R(\xi_1, \eta_1, z_0) \cdot \pi r^2 = \left(\frac{\partial P}{\partial x} + \frac{\partial R}{\partial z} \right) \Big|_{(\xi_2, \eta_2, \zeta_2)} \cdot \frac{2}{3} \pi r^2$$

$$\Rightarrow -R(\xi_1, \eta_1, z_0) = \left(\frac{\partial P}{\partial x} + \frac{\partial R}{\partial z} \right) \Big|_{(\xi_2, \eta_2, \zeta_2)} \cdot \frac{2}{3} r$$

$$\text{令 } r \rightarrow 0^+$$

$$\Rightarrow R(x_0, y_0, z_0) = 0$$

$$\text{由 } (x_0, y_0, z_0) \text{ 的任意性} \Rightarrow R(x, y, z) = 0, \forall (x, y, z) \in \mathbb{R}^3$$

$$\Rightarrow \iiint_V \frac{\partial P}{\partial x} dx dy dz = 0$$

由三重积分中值定理,

$$\exists (\xi_3, \eta_3, \zeta_3) \in V, \text{ s.t. } \frac{\partial P}{\partial x} \Big|_{(\xi_3, \eta_3, \zeta_3)} \cdot \frac{2}{3} \pi r^3 = 0$$

$$\Rightarrow \frac{\partial P}{\partial x} \Big|_{(\xi_3, \eta_3, \zeta_3)} = 0$$

$$\text{令 } r \rightarrow 0^+ \quad (\xi_3, \eta_3, \zeta_3) \rightarrow (x_0, y_0, z_0)$$

$$\Rightarrow \frac{\partial P}{\partial x}(x_0, y_0, z_0) = 0$$

$$\text{由 } (x_0, y_0, z_0) \text{ 的任意性} \Rightarrow \frac{\partial P}{\partial x}(x, y, z) = 0, \forall (x, y, z) \in \mathbb{R}^3$$

六. 保守场

(一) 势函数.

$$1. (x^2 - 2yz) dx + (y^2 - 2xz) dy + (z^2 - 2xy) dz$$

$$= d\left(\frac{1}{3}x^3 + \frac{1}{3}y^3 + \frac{1}{3}z^3 - 2xyz\right)$$

$\Rightarrow \vec{v}$ 是有势场

$$\text{势函数 } \varphi(x, y, z) = \frac{1}{3}x^3 + \frac{1}{3}y^3 + \frac{1}{3}z^3 - 2xyz + C$$

2. 积分与路径无关

$$\Rightarrow 0 = \nabla \times \vec{v} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ yf(x) & f(x)+zg(y) & g(y) \end{vmatrix}$$

$$= (g'(y) - g(y), 0, f'(x) - f(x))$$

$$\Rightarrow g'(y) = g(y) \Rightarrow g(y) = A e^y \xrightarrow{g(0)=1} e^y$$

$$f'(x) = f(x) \Rightarrow f(x) = B e^x \xrightarrow{f(0)=1} e^x$$

$$ye^x dx + (e^x + ze^y) dy + e^y dz = d(ye^x + ze^y)$$

$$\text{势函数 } \varphi(x, y, z) = ye^x + ze^y + C$$

$$3. \vec{0} = \nabla \times \vec{v} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xz & 2yf(z) & x^2 + 2yz - 1 \end{vmatrix}$$

$$= (4yz - 2yf'(z), 0, 0)$$

$$\Rightarrow 4yz - 2yf'(z) = 0$$

$$\Rightarrow f'(z) = 2z \quad f(0) = 0$$

$$\Rightarrow f(z) = z^2$$

$$2xz dx + 2yz^2 dy + (x^2 + 2yz^2 - 1) dz = d(x^2z + y^2z^2 - z)$$

$\vec{v}$ 的一个势函数为

$$\varphi(x, y, z) = x^2z + y^2z^2 - z$$

$$4. 0 = \nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ yf(x) & f(x)+ze^y & e^y \end{vmatrix}$$

$$= (0, 0, f'(x) - f(x))$$

$$f'(x) = f(x), \quad f(0) = 1$$

$$\Rightarrow f(x) = e^x$$

$$ye^x dx + (e^x + ze^y) dy + e^y dz = d(ye^x + ze^y)$$

$$\text{势函数 } \varphi(x, y, z) = ye^x + ze^y + C$$

$$5. \left(1 - \frac{y}{x} + \frac{y}{z}\right) dx + \left(\frac{x}{z} + \frac{x}{y^2}\right) dy - \frac{xy}{z^2} dz$$

$$= d\left(x - \frac{x}{y} + \frac{xy}{z}\right)$$

$$\text{势函数 } \varphi(x, y, z) = x - \frac{x}{y} + \frac{xy}{z} + C$$

一个势函数取 $C=0$

$$\varphi(x, y, z) = x - \frac{x}{y} + \frac{xy}{z}$$

$$6. yz(2x+y+z) dx + zx(x+2y+z) dy + xy(x+y+2z) dz$$

$$= d(x^2yz + xy^2z + xyz^2)$$

$$\text{势函数 } \varphi(x, y, z) = xyz(x+y+z) + C$$

7. 积分与路径无关

$$y\varphi'(x) - 2xy = 0$$

$$\Rightarrow \varphi'(x) = 2x \quad \varphi(0) = 0$$

$$\Rightarrow \varphi(x) = x^2$$

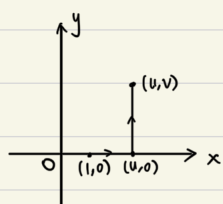
$$\int_{(0,0)}^{(1,1)} xy^2 dx + x^2y dy = \int_{(0,0)}^{(1,1)} d\left(\frac{1}{2}x^2y^2\right)$$

$$= \frac{1}{2} - 0$$

$$= \frac{1}{2}$$

$$8. (1) \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 0 \text{ 且 } D \text{ 为单连通区域.}$$

因此, $\vec{v}$ 是保守场.



$$u > 0.$$

$$\varphi(u, v) = \int_{(1,0)}^{(u,v)} \frac{-y}{a^2x^2+b^2y^2} dx + \frac{x}{a^2x^2+b^2y^2} dy$$

$$= \int_0^v \frac{u}{a^2u^2+b^2y^2} dy$$

$$= \frac{1}{ab} \arctan\left(\frac{by}{au}\right)$$

$$(2) \text{ 取 } L: (ax)^2 + (by)^2 = 1. \text{ 则 } \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

$$\oint_L \vec{v} \cdot d\vec{r} = \oint_L \frac{-y}{a^2x^2+b^2y^2} dx + \frac{x}{a^2x^2+b^2y^2} dy$$

$$= \oint_L -y dx + x dy$$

$$= 2\sigma(D).$$

$$= 2\pi \frac{1}{a} \frac{1}{b}$$

$$= \frac{2\pi}{ab} \neq 0.$$

故 $\vec{v}$ 不是区域 $D_2 = \{(x, y) \in \mathbb{R}^2 \mid x+y > 0\}$ 上的保守场.

$$9. (1) 0 = \nabla \times \vec{V} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f(y, z) & xz & xy \end{vmatrix}$$

$$= (0, \frac{\partial f}{\partial z} - y, z - \frac{\partial f}{\partial y})$$

$$\Rightarrow \frac{\partial f}{\partial z} = y$$

$$\frac{\partial f}{\partial y} = z$$

$$\Rightarrow f(y, z) = yz + C$$

$$(2) f(0,0) = 0 \Rightarrow f(y,z) = yz$$

$$yz dx + xz dy + xy dz = d(xyz)$$

$$\text{势函数 } u(x, y, z) = xyz$$

$$(3) \nabla u = (yz, zx, xy) \Big|_M = (1, 1, 1)$$

$$\frac{\partial u}{\partial n} = \nabla u \cdot \vec{n}$$

$$\left(\frac{\partial u}{\partial n}\right)_{\max} = |\nabla u| = \sqrt{3}$$

$$\left(\frac{\partial u}{\partial n}\right)_{\min} = -|\nabla u| = -\sqrt{3}$$

(=) 曲线积分与路径无关.

$$1. (x^2-yz) dx + (y^2-zx) dy + (z^2-xy) dz$$

$$= d\left(\frac{1}{3}x^3 + \frac{1}{3}y^3 + \frac{1}{3}z^3 - xyz\right)$$

$$\int_{(0,0,0)}^{(1,1,1)} (x^2-yz) dx + (y^2-zx) dy + (z^2-xy) dz$$

$$= \frac{1}{3}x^3 + \frac{1}{3}y^3 + \frac{1}{3}z^3 - xyz \Big|_{(0,0,0)}^{(1,1,1)}$$

$$= 0$$

$$2. (1) \frac{\partial}{\partial x} \left(\frac{\varphi(x)}{x^4+y^2} \right) - \frac{\partial}{\partial y} \left(\frac{2xy}{x^4+y^2} \right) = 0$$

$$\frac{\varphi'(x)(x^4+y^2) - \varphi(x) \cdot 4x^3}{(x^4+y^2)^2} - \frac{2x(x^4+y^2) - 2xy \cdot 2y}{(x^4+y^2)^2} = 0$$

$$\Rightarrow y^2(\varphi'(x) + 2x) + x^4\varphi'(x) - 4x^3\varphi(x) - 2x^5 = 0$$

$$\Rightarrow y^2(\varphi'(x) + 2x) + x^3(x\varphi'(x) - 4\varphi(x) - 2x^2) = 0$$

由 y 的任意性 $\Rightarrow \varphi'(x) + 2x = 0$

$$x\varphi'(x) - 4\varphi(x) - 2x^2 = 0$$

$$\Rightarrow \varphi(x) = -x^2$$

$$(2) L_\varepsilon: x^2 + y^2 = \varepsilon^2, \quad \varepsilon > 0 \text{ 充分小.}$$

$$\oint_{L+\bar{L}} \frac{2xy dx - x^2 dy}{x^4+y^2} = \iint_S 0 dx dy = 0$$

$$\Rightarrow \oint_C \frac{2xy dx - x^2 dy}{x^4+y^2} = \oint_{L_\varepsilon} \frac{2xy dx - x^2 dy}{x^4+y^2}$$

$$= \frac{1}{\varepsilon^2} \oint_{L_\varepsilon} 2xy dx - x^2 dy$$

$$= \frac{1}{\varepsilon^2} \iint_{S_\varepsilon} (-2x - 2y) dx dy$$

$$= 0$$

$$3. (1) \text{ 用 } (2) \varphi(x) = -x \quad (3) \oint_C \frac{y dx - x dy}{x^2 + 4y^2} = -\pi$$

t. Fourier 分析

(一) Fourier 级数

$$1. f(x) = \begin{cases} \pi+x & -\pi \leq x < 0 \\ \pi-x & 0 \leq x < \pi \end{cases}$$

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{2}{\pi} \int_0^{\pi} f(x) dx = \frac{2}{\pi} \int_0^{\pi} (\pi-x) dx = \pi$$

$$\begin{aligned} a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx = \frac{2}{\pi} \int_0^{\pi} (\pi-x) \cos nx \\ &= \frac{2}{\pi} \left[\frac{1}{n} (\pi-x) \sin nx - \frac{1}{n^2} \cos nx \right] \Big|_0^{\pi} \\ &= \frac{2[-(-1)^n]}{n^2 \pi} \end{aligned}$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx = 0$$

$$f(x) = \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{2[-(-1)^n]}{n^2 \pi} \cos nx$$

$$x=0 \Rightarrow \pi = \frac{\pi}{2} + \frac{1}{\pi} \sum_{n=1}^{\infty} \frac{4}{(2n-1)^2}$$

$$\Rightarrow \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} = \frac{\pi^2}{8}$$

$$\begin{aligned} \sum_{n=1}^{\infty} \frac{1}{n^2} &= \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} + \sum_{n=1}^{\infty} \frac{1}{(2n)^2} \\ &= \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} + \frac{1}{4} \sum_{n=1}^{\infty} \frac{1}{n^2} \end{aligned}$$

$$\Rightarrow \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{4}{3} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} = \frac{4}{3} \times \frac{\pi^2}{8} = \frac{\pi^2}{6}$$

逐项积分

$$\int_0^x (\pi-t) dt = \frac{\pi}{2} x + \sum_{n=1}^{\infty} \int_0^x \frac{4}{\pi(2n-1)^2} \cos(2n-1)t dt$$

$$\Rightarrow \pi x - \frac{1}{2} x^2 = \frac{\pi}{2} x + \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^3} \sin(2n-1)x$$

$$\text{令 } x = \frac{\pi}{2}$$

$$\Rightarrow \frac{3}{8} \pi^2 = \frac{\pi^2}{4} + \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n-1)^3}$$

$$\Rightarrow \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n-1)^3} = \frac{\pi^3}{32}$$

Parseval 等式

$$\frac{\pi^2}{2} + \sum_{n=1}^{\infty} \left(\frac{4}{\pi(2n-1)^2} \right)^2 = \frac{1}{\pi} \int_{-\pi}^{\pi} f^2(x) dx = \frac{2}{\pi} \int_0^{\pi} (\pi-x)^2 dx = \frac{2\pi^2}{3}$$

$$\Rightarrow \sum_{n=1}^{\infty} \frac{1}{(2n-1)^4} = \frac{\pi^4}{96}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^4} = \sum_{n=1}^{\infty} \frac{1}{(2n-1)^4} + \sum_{n=1}^{\infty} \frac{1}{(2n)^4}$$

$$\Rightarrow \frac{15}{16} \sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{96}$$

$$\Rightarrow \sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{16}{15} \times \frac{\pi^4}{96} = \frac{\pi^4}{90}$$

$$2. f(x) = 1-x^2$$

偶延拓

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{2}{\pi} \int_0^{\pi} (1-x^2) dx = 2\left(1-\frac{\pi^2}{3}\right)$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx = \frac{2}{\pi} \int_0^{\pi} (1-x^2) \cos nx dx = \frac{4(-1)^{n-1}}{n^2}$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx = 0$$

$f(x)$ 在 $[-\pi, \pi]$ 上连续且逐段光滑. $f_0(\pi) = f_0(-\pi)$

$$\therefore f(x) \sim 1 - \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^2} \cos nx = \begin{cases} f_0(x) & -\pi \leq x \leq \pi \\ f_0(x-2k\pi) & -\pi+2k\pi \leq x \leq \pi+2k\pi \end{cases}$$

$$\text{令 } x=0 \Rightarrow 1 - \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^2} = 1$$

$$\Rightarrow \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^2} = \frac{\pi^2}{12}$$

Parseval 等式

$$2\left(1-\frac{\pi^2}{3}\right)^2 + 16 \sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{1}{\pi} \int_{-\pi}^{\pi} (1-x^2)^2 dx$$

$$= \frac{2}{\pi} \int_0^{\pi} (1-x^2)^2 dx$$

$$= \frac{2}{\pi} \int_0^{\pi} (1-2x^2+x^4) dx$$

$$= \frac{2}{\pi} \left(\pi - 2 \cdot \frac{\pi^3}{3} + \frac{\pi^5}{5} \right)$$

$$2\left(1-\frac{\pi^2}{3} + \frac{\pi^4}{9}\right) + 16 \sum_{n=1}^{\infty} \frac{1}{n^4} = 2 - \frac{4\pi^2}{3} + \frac{2\pi^4}{5}$$

$$\Rightarrow \sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}$$

$$3. f(x) = x^2$$

偶延拓

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{2}{\pi} \int_0^{\pi} x^2 dx = \frac{2\pi^2}{3}$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx = \frac{4(-1)^{n-1}}{n^2}$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx = 0$$

$f(x)$ 在 $[-\pi, \pi]$ 上连续且逐段光滑. $f_0(\pi) = f_0(-\pi)$

$$f(x) \sim \frac{\pi^2}{3} + \sum_{n=1}^{\infty} 4 \frac{(-1)^{n-1}}{n^2} \cos nx = \begin{cases} f_0(x) & -\pi \leq x \leq \pi \\ f_0(x-2k\pi) & -\pi+2k\pi \leq x \leq \pi+2k\pi \end{cases}$$

4. (1) $f(x) = \begin{cases} 1 & 1 < x \leq 2 \\ 3-x & 2 < x \leq 3 \end{cases}$

$$a_n = \frac{1}{\pi} \int_1^3 f(x) \cos \frac{n\pi x}{\pi} dx$$

$$= \int_1^2 \cos n\pi x dx + \int_2^3 (3-x) \cos n\pi x dx$$

$$= \frac{1}{n^2 \pi^2} [1 - (-1)^n]$$

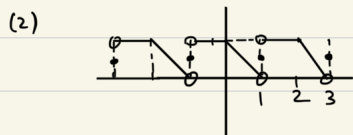
$$b_n = \frac{1}{\pi} \int_1^3 f(x) \sin \frac{n\pi x}{\pi} dx$$

$$= \int_1^2 \sin n\pi x dx + \int_2^3 (3-x) \sin n\pi x dx$$

$$= \frac{(-1)^n}{n\pi}$$

$$f(x) \sim \frac{3}{4} + \sum_{n=1}^{\infty} \left[\frac{1}{n^2 \pi^2} [1 - (-1)^n] \cos n\pi x + \frac{(-1)^n}{n\pi} \sin n\pi x \right]$$

$$= \begin{cases} \frac{1}{2} & x=1, 3 \\ 1 & 1 < x \leq 2 \\ 3-x & 2 < x < 3 \end{cases}$$



(3) $x=0$.

$$1 = \frac{3}{4} + \sum_{n=1}^{\infty} \frac{2}{(2n-1)^2 \pi^2}$$

$$\Rightarrow \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} = \frac{\pi^2}{8}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} + \sum_{n=1}^{\infty} \frac{1}{(2n)^2}$$

$$\Rightarrow \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{4}{3} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} = \frac{\pi^2}{6}$$

5. (1) $\ell=1$ $f(x) = \begin{cases} 1 & -1 < x \leq 0 \\ 1-x & 0 < x \leq 1 \end{cases}$

$$a_0 = \frac{1}{\pi} \int_{-1}^1 f(x) dx = \int_{-1}^0 1 dx + \int_0^1 (1-x) dx = 1 + 1 - \frac{1}{2} = \frac{3}{2}$$

$$a_n = \frac{1}{\pi} \int_{-1}^1 f(x) \cos \frac{n\pi x}{\pi} dx = \frac{1 - (-1)^n}{n^2 \pi^2}$$

$$b_n = \frac{1}{\pi} \int_{-1}^1 f(x) \sin \frac{n\pi x}{\pi} dx = \frac{(-1)^n}{n\pi}$$

$$f(x) \sim \frac{3}{4} + \sum_{n=1}^{\infty} \left(\frac{(-1)^{n+1} + 1}{n^2 \pi^2} \cos n\pi x + \frac{(-1)^n}{n\pi} \sin n\pi x \right)$$

$$= \begin{cases} 1 & -1 < x \leq 0 \\ 1-x & 0 < x < 1 \\ \frac{1}{2} & x=-1, 1 \end{cases}$$

(2) Parseval 等式

$$\frac{1}{2} \left(\frac{3}{2} \right)^2 + \sum_{n=1}^{\infty} \frac{4}{(2n-1)^4 \pi^4} + \sum_{n=1}^{\infty} \frac{1}{n^2 \pi^2} = \frac{1}{\pi} \int_{-1}^1 f^2(x) dx = \frac{4}{3}$$

(3) $x=0 \Rightarrow \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} = \frac{\pi^2}{8} \quad \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$

$$\frac{5}{24} = \frac{4}{\pi^4} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^4} + \frac{1}{6} \Rightarrow \sum_{n=1}^{\infty} \frac{1}{(2n-1)^4} = \frac{\pi^4}{96}$$

$$\Rightarrow \sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}$$

6. $f(x)=x$ ~~周期延拓~~

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{2}{\pi} \int_0^{\pi} x dx = \pi$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx = \frac{2}{\pi} \int_0^{\pi} x \cos nx dx$$

$$= \frac{2}{\pi} \frac{(-1)^n - 1}{n^2}$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx = 0$$

$$f(x) \sim \frac{\pi}{2} - \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} \cos(2n-1)x = x \quad 0 \leq x \leq \pi$$

$$\sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} = \frac{\pi^2}{8}$$

$$\Rightarrow \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} = \frac{\pi^2}{8}$$

7. $f(x) = \begin{cases} 0 & -\pi < x \leq -\frac{\pi}{2} \\ 1 & -\frac{\pi}{2} < x < \frac{\pi}{2} \\ 0 & \frac{\pi}{2} \leq x < \pi \end{cases}$

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{\pi} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 1 dx = 1$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx = \frac{2}{\pi} \int_0^{\frac{\pi}{2}} \cos nx dx$$

$$= \frac{2}{n\pi} \sin \frac{n\pi}{2}$$

$$= \begin{cases} 0 & n=4k, 4k+2 \\ \frac{2}{(4k+1)\pi} & n=4k+1 \\ -\frac{2}{(4k+3)\pi} & n=4k+3 \end{cases}$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx = 0$$

$$f(x) \sim \frac{1}{2} + \sum_{n=0}^{\infty} \frac{2}{\pi} \frac{(-1)^n}{2n+1} \cos(2n+1)x = \begin{cases} 1 & |x| < \frac{\pi}{2} \\ \frac{1}{2} & |x| = \frac{\pi}{2} \\ 0 & \frac{\pi}{2} < |x| \leq \pi \end{cases}$$

$$\sum_{n=0}^{\infty} \frac{1}{(2n+1)^2} = \frac{\pi^2}{8}$$

$$\Rightarrow \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} = \frac{\pi}{4}$$

Parseval 等式

$$\frac{1}{2} + \frac{4}{\pi^2} \sum_{n=0}^{\infty} \frac{1}{(2n+1)^2} = \frac{1}{\pi} \int_{-\pi}^{\pi} f^2(x) dx = \frac{2}{\pi} \int_0^{\frac{\pi}{2}} 1 dx = 1$$

$$\Rightarrow \sum_{n=0}^{\infty} \frac{1}{(2n+1)^2} = \frac{\pi^2}{8}$$

$$8. f(x) = \frac{\pi}{2} - x \quad x \in [0, \pi]$$

偶延拓.

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{2}{\pi} \int_0^{\pi} (\frac{\pi}{2} - x) dx = 0$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx = \frac{2}{\pi} \frac{1 - (-1)^n}{n^2}$$

$$f(x) \sim \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{\cos(2n-1)x}{(2n-1)^2} = \frac{\pi}{2} - x \quad x \in [0, \pi]$$

$$\sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} = \frac{\pi^2}{8}$$

$$\Rightarrow \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} = \frac{\pi^2}{8}$$

Parseval 等式

$$\sum_{n=1}^{\infty} \frac{16}{\pi^2} \frac{1}{(2n-1)^4} = \frac{1}{\pi} \int_{-\pi}^{\pi} f^2(x) dx = \frac{2}{\pi} \int_0^{\pi} (\frac{\pi}{2} - x)^2 dx$$

$$= \frac{2}{\pi} \left(\frac{\pi^2}{3} \cdot \pi - \pi \frac{\pi^2}{2} + \frac{\pi^3}{3} \right)$$

$$= 2 \left(\frac{1}{4} - \frac{1}{2} + \frac{1}{3} \right) \pi^2$$

$$= \frac{\pi^2}{6}$$

$$\Rightarrow \sum_{n=1}^{\infty} \frac{1}{(2n-1)^4} = \frac{\pi^4}{96}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^4} = \sum_{n=1}^{\infty} \frac{1}{(2n-1)^4} + \frac{1}{16} \sum_{n=1}^{\infty} \frac{1}{n^4}$$

$$\Rightarrow \frac{15}{16} \sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{96}$$

$$\Rightarrow \sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{16}{15} \times \frac{\pi^4}{96} = \frac{\pi^4}{90}$$

$$9. i\text{证}: a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx$$

$$x = t + \frac{\pi}{n}$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t + \frac{\pi}{n}) \cos(nt + \pi) dt$$

$$= -\frac{1}{\pi} \int_{-\pi}^{\pi} f(x + \frac{\pi}{n}) \cos nx dx$$

$$\Rightarrow a_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} (f(x) - f(x + \frac{\pi}{n})) \cos nx dx$$

$$|a_n| \leq \frac{1}{2\pi} \int_{-\pi}^{\pi} |f(x) - f(x + \frac{\pi}{n})| |\cos nx| dx$$

$$\leq \frac{1}{2\pi} \int_{-\pi}^{\pi} (\frac{\pi}{n})^{\alpha} |\cos nx| dx$$

$$= \frac{1}{2\pi} (\frac{\pi}{n})^{\alpha} \int_{-\pi}^{\pi} |\cos nx| dx$$

$$= \frac{2}{\pi} (\frac{\pi}{n})^{\alpha}$$

$$< (\frac{\pi}{n})^{\alpha}$$

$$\text{同理, } |b_n| < (\frac{\pi}{n})^{\alpha}$$

$$10. \text{证}: g(x) = \pi - x \text{ 在 } [0, 2\pi] \text{ 上 平方可积.}$$

$$\alpha_0 = \frac{1}{\pi} \int_0^{2\pi} (\pi - x) dx = 0$$

$$\alpha_n = \frac{1}{\pi} \int_0^{2\pi} (\pi - x) \cos nx dx = 0$$

$$\beta_n = \frac{1}{\pi} \int_0^{2\pi} (\pi - x) \sin nx dx = \frac{1}{\pi} \frac{2\pi}{n} = \frac{2}{n}$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx dx \quad f(x) \text{ 奇.}$$

$$a_n = 0$$

Parseval 等式

$$\frac{1}{\pi} \int_0^{2\pi} f(x)(\pi - x) dx = \frac{\alpha_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2)$$

$$= 0 + \sum_{n=1}^{\infty} (0 + b_n^2 \cdot \frac{2}{n})$$

$$\Rightarrow \sum_{n=1}^{\infty} \frac{b_n^2}{n} = \frac{1}{2\pi} \int_0^{2\pi} f(x)(\pi - x) dx$$

(二) Fourier 变换.

$$1. f(x) = \begin{cases} 1 & x \in [-\pi, \pi] \\ 0 & \text{其他} \end{cases}$$

$$F(\lambda) = \int_{-\pi}^{\pi} f(x) e^{-i\lambda x} dx$$

$$= \int_{-\pi}^{\pi} e^{-i\lambda x} dx$$

$$= \frac{e^{-i\lambda\pi} - e^{i\lambda\pi}}{-i\lambda}$$

$$= \frac{-2i \sin \lambda\pi}{-i\lambda}$$

$$= \frac{2 \sin \lambda\pi}{\lambda}$$

$$2. f(x) = \begin{cases} e^{-2x} & x \geq 0 \\ e^{2x} & x < 0 \end{cases}$$

$$F(\lambda) = \int_{-\infty}^{+\infty} f(x) e^{-i\lambda x} dx$$

$$= \int_{-\infty}^0 e^{2x} e^{-i\lambda x} dx + \int_0^{+\infty} e^{-2x} e^{-i\lambda x} dx$$

$$= \frac{1}{2-i\lambda} + \frac{1}{2+i\lambda}$$

$$= \frac{4}{4+\lambda^2}$$

八. 反常积分、含参变量积分

(一) 反常积分收敛性

$$1. \int_1^{+\infty} \frac{e^{\sin x} \cos x}{x^p} \left(1 + \frac{1}{x}\right)^x dx \quad (p > 0)$$

$$\text{令 } f(x) = \left(1 + \frac{1}{x}\right)^x \quad g(x) = x \ln \left(1 + \frac{1}{x}\right)$$

$$h(t) = \frac{\ln(1+t)}{t} \quad 0 < t \leq 1$$

$$h'(t) = \frac{\frac{1}{1+t} - \ln(1+t)}{t^2} < 0 \quad h(t) \downarrow$$

$$\varphi(t) = \frac{t}{1+t} - \ln(1+t)$$

$$\varphi'(t) = \frac{1}{(1+t)^2} - \frac{1}{1+t} < 0.$$

$f(x)$ 在 $x \geq 1$ 时 $\uparrow$, $f(x)$ 的值域为 $[2, e)$

(1) 当 $p > 1$ 时,

$$\left| \frac{e^{\sin x} \cos x}{x^p} \left(1 + \frac{1}{x}\right)^x \right| \leq \frac{e^2}{x^p} \quad (x \geq 1).$$

$$\int_1^{+\infty} \frac{e^2}{x^p} dx \text{ 收敛.}$$

由比较判别法 $\Rightarrow$ 原积分绝对收敛.

(2) 当 $0 < p \leq 1$ 时,

$$\forall A > 1, \left| \int_1^A e^{\sin x} \cos x \right| = \left| e^{\sin x} \right|_1^A \leq 2e$$

当 $x \geq 1$ 时, $\frac{1}{x^p} \downarrow$ 趋于 0.

$$\text{由 Dirichlet 判别法} \Rightarrow \int_1^{+\infty} \frac{e^{\sin x} \cos x}{x^p} dx \text{ 收敛.}$$

$f(x) = \left(1 + \frac{1}{x}\right)^x \uparrow$ 且有界.

由 Abel 判别法 $\Rightarrow$ 原积分收敛.

$$\left| \frac{e^{\sin x} \cos x}{x^p} \left(1 + \frac{1}{x}\right)^p \right| \geq \frac{2}{e} \frac{\cos^2 x}{x^p} = \frac{1 + \cos 2x}{e x^p}$$

$$\int_1^{+\infty} \frac{1}{e x^p} dx \text{ 发散.} \quad \int_1^{+\infty} \frac{\cos 2x}{e x^p} dx \text{ 收敛.}$$

$$\text{由比较判别法} \Rightarrow \int_1^{+\infty} \left| \frac{e^{\sin x} \cos x}{x^p} \left(1 + \frac{1}{x}\right)^x \right| dx \text{ 发散.}$$

因此, 当 $0 < p \leq 1$ 时, 原积分条件收敛.

(二) 含参变量积分的性质.

1. $\frac{1}{1+x^2}$ 在 $[0, 2] \times [0, 1]$ 上连续

由含参变量积分的连续性

$$\Rightarrow \lim_{\alpha \rightarrow 0} \int_0^{1+\alpha} \frac{1}{1+x^2} dx = \int_0^1 \frac{1}{1+x^2} dx = \frac{\pi}{4}$$

$$2. G'(\alpha) = \int_0^\alpha \frac{1}{1+\alpha x} dx + 1 \cdot \frac{\ln(1+\alpha^2)}{\alpha}$$

$$= \frac{1}{\alpha} \ln(1+\alpha x) \Big|_0^\alpha + \frac{1}{\alpha} \ln(1+\alpha^2)$$

$$= \frac{2}{\alpha} \ln(1+\alpha^2)$$

$$F'(\alpha) = \int_0^{\cos \alpha} \frac{e^{\alpha \sqrt{1-x^2}}}{\sqrt{1-x^2}} dx + e^{\alpha \sqrt{1-x^2}} \Big|_0^{\cos \alpha} \cdot (-\sin \alpha)$$

3. $f(x, \alpha) = e^{x^2 + x\alpha}$ 在 $[-1, 1] \times [-\frac{\pi}{2}, \frac{\pi}{2}]$ 连续, $\frac{\partial f}{\partial \alpha}$ 在 $[-1, 1] \times [-\frac{\pi}{2}, \frac{\pi}{2}]$ 连续

由含参变量积分的可微性

$$I'(\alpha) = \int_{\sin \alpha}^{\cos \alpha} e^{x^2 + x\alpha} \cdot x dx + (-\sin \alpha) e^{\cos^2 \alpha + \cos \alpha \cdot \alpha} - \cos \alpha e^{\sin^2 \alpha + \sin \alpha \cdot \alpha}$$

$$I'(0) = \int_0^1 e^{x^2} x dx - 1$$

$$= \frac{1}{2} e^{x^2} \Big|_0^1 - 1$$

$$= \frac{1}{2} (e - 1) - 1$$

$$= \frac{1}{2} (e - 3)$$

$$4. F'(\alpha) = \int_{\sin \alpha}^{\cos \alpha} \left(f'_1(x+\alpha, x-\alpha) - f'_1(x+\alpha, x-\alpha) \right) dx$$

$$+ (-\sin \alpha) f(\cos \alpha + \alpha, \cos \alpha - \alpha)$$

$$- \cos \alpha f(\sin \alpha + \alpha, \sin \alpha - \alpha)$$

$$5. g'(x) = \int_{2^x}^{4^x} \left(-t^2 e^{-xt^2} - \sin(xt)^2 \cdot 2xt \cdot t \right) dt$$

$$+ 3 \cos^2 x (-\sin x) \left(e^{-x \cos^2 x} + \cos(x^2 \cos^6 x) \right)$$

$$- 2^x \ln 2 \left(e^{-x \cdot 4^x} + \cos(x^2 \cdot 4^x) \right)$$

$$g'(0) = 0 + 0 - \ln 2 (1+1) = -2 \ln 2$$

$$6. F'(x) = \int_0^x u f(u^2 + x^2) du$$

$$F''(x) = \int_0^x 2ux f'(u^2 + x^2) du + x f(2x^2) - 0$$

$$= x (f(2x^2) - f(x^2)) + x f(2x^2)$$

$$= 2x f(2x^2) - x f(x^2)$$

(二) 含参变量反常积分的一致收敛性.

$$1. I(\alpha) = \int_1^{+\infty} \frac{\sin x}{x^\alpha} dx$$

(1) $\alpha \in [\alpha_0, +\infty)$, $\alpha_0 > 0$

$$\forall A > 1, \left| \int_1^A \sin x dx \right| = |\cos 1 - \cos A| \leq 2 \text{ 一致有界}$$

$\frac{1}{x^\alpha}$ 单调 $\downarrow$ - 一致趋于 0.

由 Dirichlet 判别法.

$\Rightarrow I(\alpha)$ 关于 α 在 $[\alpha_0, +\infty)$ 上一致收敛.

(2) 不一致收敛.

(反证法) 假设积分在 $(0, +\infty)$ 上一致收敛.

由 Cauchy 收敛准则

取 $\varepsilon_0 = 1$. $\exists A_0 > 1$, $\forall A_2 > A_1 > A_0$, $\alpha > 0$,

$$\left| \int_{A_1}^{A_2} \frac{\sin x}{x^\alpha} dx \right| < 1$$

$$\text{令 } A_1 = 2n\pi, A_2 = (2n+1)\pi. \alpha = \frac{1}{2n}$$

$$[2n\pi, (2n+1)\pi] \text{ 上 } \sin x \geq 0.$$

$$\begin{aligned} x^\alpha &\leq [(2n+1)\pi]^\alpha \\ \int_{2n\pi}^{(2n+1)\pi} \frac{\sin x}{x^\alpha} dx &\geq \frac{1}{[(2n+1)\pi]^\alpha} \int_{2n\pi}^{(2n+1)\pi} \sin x dx \\ &= \frac{2}{[(2n+1)\pi]^{\frac{1}{2n}}} \end{aligned}$$

$$\lim_{n \rightarrow \infty} [(2n+1)\pi]^{\frac{1}{2n}} = 1$$

$$\text{因此, 当 } n \text{ 充分大时, } \frac{1}{[(2n+1)\pi]^{\frac{1}{2n}}} > 1$$

矛盾! 故积分在 $(0, +\infty)$ 上不一致收敛.

$$2. \int_0^{+\infty} \frac{\sin 2x}{x+\alpha} e^{-\alpha x} dx \quad \alpha \in [b, B] \quad 0 < b < B.$$

$$\left| \int_0^A \sin 2x dx \right| = \left| \frac{1 - \cos 2A}{2} \right| \leq 1 \text{ 一致有界.}$$

$\frac{1}{x+\alpha} e^{-\alpha x}$ 单调 $\downarrow$ - 一致趋于 0.

由 Dirichlet 判别法

$\Rightarrow \int_0^{+\infty} \frac{\sin 2x}{x+\alpha} e^{-\alpha x} dx$ 在 $\alpha \in [b, B]$ 上一致收敛.

(三) 利用含参变量积分的性质计算积分

$$1. f(x, u) = \ln(\sin^2 x + u^2 \cos^2 x) \text{ 和 } \frac{\partial f}{\partial u} = \frac{2u \cos^2 x}{\sin^2 x + u^2 \cos^2 x}$$

在 $[0, \frac{\pi}{2}] \times (0, +\infty)$ 上连续

由含参变量积分的可微性.

$$\begin{aligned} F'(u) &= \int_0^{\frac{\pi}{2}} \frac{2u \cos^2 x}{\sin^2 x + u^2 \cos^2 x} dx \\ &= \int_0^{\frac{\pi}{2}} \frac{2u}{t^2 + u^2} \frac{dt}{1+t^2} \\ &= \frac{2u}{1-u^2} \int_0^{+\infty} \left(\frac{1}{t^2 + u^2} - \frac{1}{t^2 + 1} \right) dt \\ &= \frac{2u}{1-u^2} \left(\frac{1}{u} \arctan \left(\frac{t}{u} \right) - \arctan t \right) \Big|_0^{+\infty} \\ &= \frac{\pi}{1+u} \end{aligned}$$

$$F(1) = 0$$

$$F(u) = F(1) + \int_1^u \frac{\pi}{1+t} dt = \pi \ln \frac{1+u}{2}$$

$$2. \frac{\arctan x}{x(1+x^2)} \text{ 和 } \frac{1}{(1+x^2)(1+\alpha^2 x^2)} \text{ 在 } [0, +\infty) \times (0, +\infty) \text{ 上连续.}$$

$$\left| \frac{\arctan x}{x(1+x^2)} \right| \leq \frac{x}{1+x^2} \quad \int_0^{+\infty} \frac{x}{1+x^2} dx \text{ 收敛}$$

由比较判别法.

$$\Rightarrow \int_0^{+\infty} \frac{\arctan x}{x(1+x^2)} dx \text{ 关于 } \alpha \text{ 在 } (0, +\infty) \text{ 上收敛.}$$

$$\left| \frac{1}{(1+x^2)(1+\alpha^2 x^2)} \right| \leq \frac{1}{1+x^2} \quad \int_0^{+\infty} \frac{1}{1+x^2} dx \text{ 收敛}$$

由 Weierstrass 判别法.

$$\Rightarrow \int_0^{+\infty} \frac{1}{(1+x^2)(1+\alpha^2 x^2)} dx \text{ 在 } [0, +\infty) \text{ 上一致收敛.}$$

由一致收敛含参变量广义积分的可微性.

$$\begin{aligned} \Rightarrow I'(\alpha) &= \int_0^{+\infty} \frac{1}{(1+x^2)(1+\alpha^2 x^2)} dx \\ &= \frac{1}{1-\alpha^2} \int_0^{+\infty} \left(\frac{1}{1+x^2} - \frac{\alpha^2}{1+\alpha^2 x^2} \right) dx \\ &= \frac{1}{1-\alpha^2} \cdot \frac{\pi}{2} (1-\alpha) \\ &= \frac{\pi}{2(1+\alpha)} \end{aligned}$$

$$I(0) = 0$$

$$\Rightarrow I(\alpha) = I(0) + \int_0^\alpha \frac{\pi}{2(1+t)} dt = \frac{\pi}{2} \ln(1+\alpha)$$

$$3. I(a) = \int_0^{+\infty} \frac{\ln(1+a^2x^2)}{1+x^2} dx \quad (a>0)$$

$$I(0) = 0.$$

当 $0 \leq a \leq a_1$ ($a_1 > 0$) 时,

$$0 \leq \frac{\ln(1+a^2x^2)}{1+x^2} \leq \frac{\ln(1+a_1^2x^2)}{1+x^2} \quad \frac{\ln(1+a_1^2x^2)}{1+x^2} = o\left(\frac{1}{x^2}\right)$$

且 $\int_0^{+\infty} \frac{\ln(1+a_1^2x^2)}{1+x^2} dx$ 收敛.

由 Weierstrass 判别法 $\Rightarrow I(a)$ 在 $0 \leq a \leq a_1$ 上一致收敛.

$\Rightarrow I(a)$ 在 $[0, +\infty)$ 上连续.

当 $a > a_0 > 0$ 时

$$0 \leq \frac{2ax^2}{(1+x^2)(1+a^2x^2)} \leq \frac{2}{a_0} \frac{1}{1+x^2} \quad x \geq 0.$$

$\int_0^{+\infty} \frac{2}{a_0} \frac{1}{1+x^2} dx$ 收敛.

由 Weierstrass 判别法 $\Rightarrow I'(a)$ 在 $a > a_0 > 0$ 上关于 a 一致收敛
且 $I'(a)$ 在 $a > 0$ 时连续.

由 Th 13.31

$$I'(a) = \int_0^{+\infty} \frac{\partial}{\partial a} \left(\frac{\ln(1+a^2x^2)}{1+x^2} \right) dx$$

$$= \int_0^{+\infty} \frac{2ax^2}{(1+x^2)(1+a^2x^2)} dx$$

$$= \frac{2a}{1-a^2} \int_0^{+\infty} \left(\frac{1}{1+a^2x^2} - \frac{1}{1+x^2} \right) dx$$

$$= \frac{2a}{1-a^2} \left(\frac{1}{a} \frac{\pi}{2} - \frac{\pi}{2} \right)$$

$$= \frac{\pi}{1+a} \quad a > 0$$

$$I(a) = \pi \ln(1+a) + C \quad a \geq 0.$$

$$I(0) = 0 \Rightarrow C = 0$$

所以, $I(a) = \pi \ln(1+a)$.

(四) 利用几个重要积分计算

$$\begin{aligned} 1. \int_0^{+\infty} \frac{\sin^2 x}{x^2} dx &= -\frac{1}{x} \sin^2 x \Big|_0^{+\infty} + \int_0^{+\infty} \frac{2 \sin x \cos x}{x} dx \\ &= \int_0^{+\infty} \frac{\sin 2x}{x} dx \\ &= \frac{\pi}{2} \end{aligned}$$

$$\begin{aligned} 2. \int_0^{+\infty} t^{\frac{1}{2}} e^{-at} dt &\stackrel{x=at}{=} \frac{1}{a^{\frac{3}{2}}} \int_0^{+\infty} x^{\frac{1}{2}} e^{-x} dx \\ &= \frac{1}{a^{\frac{3}{2}}} \Gamma\left(\frac{3}{2}\right) \\ &= \sqrt{\frac{\pi}{a}} \end{aligned}$$

$$\begin{aligned} 3. \int_0^{+\infty} \frac{x^4}{(9+x^2)^5} dx &\stackrel{x=3t}{=} \int_0^{+\infty} \frac{(3t)^4}{3^{10}(1+t^2)^5} \cdot 3 dt \\ &\stackrel{u=t^2}{=} \frac{1}{3^5} \int_0^{+\infty} \frac{u^2(1+u)^5}{2\sqrt{u}} du \\ &= \frac{1}{2 \cdot 3^5} \int_0^{+\infty} \frac{u^{\frac{3}{2}}}{(1+u)^5} du \end{aligned}$$

$$B\left(\frac{5}{2}, \frac{5}{2}\right) = \int_0^1 t^{\frac{1}{2}} (1-t)^{\frac{3}{2}} dt = \int_0^{+\infty} \frac{z^{\frac{3}{2}-1}}{(1+z)^{\frac{5}{2}+\frac{5}{2}}} dz$$

$$= \frac{1}{2 \cdot 3^5} B\left(\frac{5}{2}, \frac{5}{2}\right)$$

$$= \frac{1}{2 \cdot 3^5} \frac{\left(\Gamma\left(\frac{5}{2}\right)\right)^2}{\Gamma(5)}$$

$$= \frac{1}{2 \cdot 3^5} \frac{\left(\frac{3}{2} \cdot \frac{1}{2} \sqrt{\pi}\right)^2}{4!}$$

$$= \frac{\pi}{2^8 \cdot 3^4}$$

$$= \frac{\pi}{20736}$$

$$\begin{aligned} 4. \quad y &= (b^n - x^n)^{\frac{1}{n}} = a \left(1 - \left(\frac{x}{a}\right)^n\right)^{\frac{1}{n}} \\ t &= \frac{x}{a}. \end{aligned}$$

$$S = \int_0^a y dx = \int_0^1 a(1-t^n)^{\frac{1}{n}} \cdot a dt$$

$$= a^2 \int_0^1 (1-t^n)^{\frac{1}{n}} dt$$

$$\begin{aligned} t &= u^{\frac{1}{n}} \\ dt &= \frac{1}{n} u^{\frac{1}{n}-1} du \end{aligned} \quad \stackrel{u=t^n}{=} a^2 \int_0^1 (1-u)^{\frac{1}{n}} \cdot \frac{1}{n} u^{\frac{1}{n}-1} du$$

$$= \frac{a^2}{n} \int_0^1 u^{\frac{1}{n}-1} (1-u)^{\frac{1}{n}} du$$

$$= \frac{a^2}{n} B\left(\frac{1}{n}, \frac{1}{n}+1\right)$$

$$5. \int_0^{+\infty} e^{-x^n} dx \stackrel{t=x^n}{=} \int_0^{+\infty} e^{-t} \frac{1}{n} t^{\frac{1}{n}-1} dt$$

$$= \frac{1}{n} \int_0^{+\infty} t^{\frac{1}{n}-1} e^{-t} dt$$

$$\begin{aligned} x &= t^{\frac{1}{n}} \\ dx &= \frac{1}{n} t^{\frac{1}{n}-1} dt \end{aligned} \quad = \frac{1}{n} \Gamma\left(\frac{1}{n}\right)$$

$$= \Gamma\left(1+\frac{1}{n}\right)$$

$$\lim_{n \rightarrow \infty} \int_0^{+\infty} e^{-x^n} dx = \lim_{n \rightarrow \infty} \Gamma\left(1+\frac{1}{n}\right) = \Gamma(1) = 1$$

$$\begin{aligned}
 x &= t^{\frac{1}{4}} \\
 dx &= \frac{1}{4} t^{-\frac{3}{4}} dt \\
 6. I &= \int_0^{+\infty} \frac{1}{1+x^4} dx \stackrel{t=x^4}{=} \int_0^{+\infty} \frac{\frac{1}{4} t^{-\frac{3}{4}}}{1+t} dt \\
 &= \frac{1}{4} B\left(\frac{1}{4}, \frac{3}{4}\right) \\
 &= \frac{1}{4} \frac{\Gamma(\frac{1}{4})\Gamma(\frac{3}{4})}{\Gamma(1)} \\
 &= \frac{1}{4} \frac{\pi}{\sin \frac{\pi}{4}} \\
 &= \frac{1}{4} \cdot \sqrt{2} \pi \\
 &= \frac{\pi}{2\sqrt{2}}
 \end{aligned}$$

(五) 含参变量广义积分收敛证明及计算.

$$\begin{aligned}
 1. (1) \varphi(\alpha) &= \int_0^1 \frac{x^\alpha}{1+x^2} dx + \int_1^{+\infty} \frac{x^\alpha}{1+x^2} dx \\
 x \rightarrow 0^+ \text{ 时, } \frac{x^\alpha}{1+x^2} &\sim x^\alpha \quad \alpha > -1 \\
 x \rightarrow +\infty \text{ 时, } \frac{x^\alpha}{1+x^2} &\sim x^{\alpha-2} \quad \alpha-2 < -1 \\
 \Rightarrow -1 < \alpha < 1 \\
 (2) \varphi(\alpha) &= \int_0^{+\infty} \frac{x^\alpha}{1+x^2} dx \stackrel{t=x^2}{=} \frac{1}{2} \int_0^{+\infty} \frac{t^{\frac{\alpha}{2}-\frac{1}{2}}}{1+t} dt = \frac{1}{2} B\left(\frac{\alpha+1}{2}, \frac{1-\alpha}{2}\right) \\
 &= \frac{1}{2} \Gamma\left(\frac{\alpha+1}{2}\right) \Gamma\left(\frac{1-\alpha}{2}\right) \\
 &= \frac{1}{2} \frac{\pi}{\sin\left(\frac{\alpha+1}{2}\pi\right)} \\
 &= \frac{\pi}{2} \sec\left(\frac{\alpha}{2}\pi\right)
 \end{aligned}$$

$$\begin{aligned}
 (3) 0 < x \leq 1 \text{ 时, } \frac{x^\alpha}{1+x^2} &\leq \frac{x^{-\alpha_0}}{1+x^2} \leq x^{-\alpha_0} \\
 0 < \alpha_0 < 1 \Rightarrow -\alpha_0 > -1 \quad \int_0^1 x^{-\alpha_0} dx &\text{ 收敛} \\
 \text{由 Weierstrass 判别法.} \\
 \Rightarrow \int_0^1 \frac{x^\alpha}{1+x^2} dx &\text{ 关于 } \alpha \text{ 在 } \alpha \geq -\alpha_0 \text{ 上一致收敛.} \\
 x > 1 \text{ 时, } \frac{x^\alpha}{1+x^2} &\leq \frac{x^{\alpha_0}}{1+x^2} \leq x^{\alpha_0-2} \\
 \alpha < \alpha_0 < 1 \Rightarrow \alpha_0-2 < -1 \Rightarrow \int_1^{+\infty} x^{\alpha_0-2} dx &\text{ 收敛.} \\
 \text{由 Weierstrass 判别法} \\
 \Rightarrow \int_1^{+\infty} \frac{x^\alpha}{1+x^2} dx &\text{ 关于 } \alpha \text{ 在 } \alpha \leq \alpha_0 \text{ 上一致收敛.} \\
 \Rightarrow \varphi(\alpha) = \int_0^{+\infty} \frac{x^\alpha}{1+x^2} dx &\text{ 关于 } \alpha \text{ 在 } -\alpha_0 \leq \alpha \leq \alpha_0 \text{ 上一致收敛}
 \end{aligned}$$

$$2. \text{ 令 } I(a) = \int_0^{+\infty} \frac{\sin(ax) \sin(bx)}{x^2} dx \quad 0 \leq a < b.$$

$$I(0) = 0.$$

$$\left| \frac{\sin(ax) \sin(bx)}{x^2} \right| \leq \frac{1}{x^2} \quad (x \geq 1)$$

由 Weierstrass 判别法

$$\Rightarrow I(a) = \int_0^{+\infty} \frac{\sin(ax) \sin(bx)}{x^2} dx \text{ 关于 } a \text{ 在 } [0, b] \text{ 上一致收敛.}$$

由一致收敛含参变量反常积分的连续性

$$\Rightarrow I(a) \text{ 在 } [0, b] \text{ 上连续.} \quad I(0) = 0.$$

$$\begin{aligned}
 \frac{\partial}{\partial a} \left(\frac{\sin(ax) \sin(bx)}{x^2} \right) &= \frac{\cos(ax) \sin(bx)}{x} \\
 &= \frac{1}{2} \left(\frac{\sin(a+b)x}{x} - \frac{\sin(a-b)x}{x} \right)
 \end{aligned}$$

$$\begin{aligned}
 \int_0^A \sin(a+b)x dx &\leq \frac{2}{a+b} \leq \frac{2}{b} \\
 \int_0^A \sin(b-a)x dx &\leq \frac{2}{b-a} \leq \frac{2}{b-a_0} \quad a < a_0 < b \\
 \frac{1}{x} \text{ 单调 } \downarrow, \text{ 当 } x \rightarrow +\infty \text{ 时, 一致趋于 } 0.
 \end{aligned}$$

由 Dirichlet 判别法.

$$\Rightarrow \int_0^{+\infty} \frac{\partial}{\partial a} \left(\frac{\sin(ax) \sin(bx)}{x^2} \right) dx \text{ 关于 } a \text{ 在 } [0, b] \text{ 上内闭一致收敛}$$

由一致收敛含参变量反常积分的可微性

$$\begin{aligned}
 \Rightarrow I'(a) &= \int_0^{+\infty} \frac{1}{2} \left(\frac{\sin(a+b)x}{x} - \frac{\sin(a-b)x}{x} \right) dx \quad (0 \leq a < b) \\
 &= \frac{1}{2} \left(\frac{\pi}{2} + \frac{\pi}{2} \right) \\
 &= \frac{\pi}{2}
 \end{aligned}$$

$$I(a) = I(0) + \int_0^a \frac{\pi}{2} da = \frac{\pi}{2} a$$

$$3. (1) \forall T > 0, \text{ 当 } 0 \leq t \leq T, x \geq 0 \text{ 时.}$$

$$0 \leq \frac{\ln(1+tx)}{1+x^2} \leq \frac{\ln(1+Tx)}{1+x^2}$$

$$\int_0^{+\infty} \frac{\ln(1+Tx)}{1+x^2} dx \text{ 收敛 由 Weierstrass 判别法}$$

$$\Rightarrow \int_0^{+\infty} \frac{\ln(1+tx)}{1+x^2} dx \text{ 关于 } t \text{ 在 } [0, T] \text{ 上一致收敛}$$

$$\Rightarrow \varphi(t) \text{ 关于 } t \text{ 在 } t \geq 0 \text{ 上连续. } \varphi(0) = 0.$$

$$\text{当 } t \geq t_0 > 0 \text{ 时, } 0 \leq \frac{1}{(1+x)(1+tx)} \leq \frac{1}{(1+x)(1+t_0x)}$$

$$x \rightarrow +\infty \text{ 时, } \frac{1}{(1+x)(1+t_0x)} \sim \frac{1}{t_0 x^2}$$

$$\int_0^{+\infty} \frac{1}{(1+x)(1+t_0x)} dx \text{ 收敛}$$

由 Weierstrass 判别法

$$\Rightarrow \int_0^{+\infty} \frac{1}{(1+t)(1+tx)} dx \text{ 关于 } t \text{ 在 } t \geq t_0 > 0 \text{ 上一致收敛}$$

$$\text{即 } \int_0^{+\infty} \frac{1}{(1+t)(1+tx)} dx \text{ 关于 } t \text{ 在 } (0, +\infty) \text{ 内闭一致收敛}$$

由一致收敛含参变量反常积分的可微性.

$$\Rightarrow \varphi'(t) = \int_0^{+\infty} \frac{1}{(1+t)(1+tx)} dx$$

$$(2) \text{ 当 } t > 0 \text{ 且 } t \neq 1 \text{ 时.}$$

$$\varphi'(t) = \int_1^{+\infty} \left(\frac{1}{1+x} \frac{1}{1-t} + \frac{t}{t-1} \frac{1}{1+tx} \right) dx$$

$$= \frac{1}{1-t} \ln \frac{1+x}{1+tx} \Big|_1^{+\infty}$$

$$= \begin{cases} -\frac{\ln t}{1-t} & t > 0 \text{ 且 } t \neq 1 \\ 1 & t = 1 \end{cases}$$

$$\varphi(0) = 0$$

当 $0 < b < 1$ 时

$$\begin{aligned}\varphi(b) &= \int_0^b \varphi'(t) dt = \int_0^b \frac{-\ln t}{1-t} dt = \int_0^b \frac{-\ln(1-x)}{x} dx \\ &= \int_0^b \sum_{n=1}^{\infty} \frac{t^{n-1}}{n} dt = \sum_{n=1}^{\infty} \frac{1}{n} \int_0^b t^{n-1} dt \\ &= \sum_{n=1}^{\infty} \frac{1}{n^2} b^n \quad \text{--- } -\ln(1-x) = \sum_{n=1}^{\infty} \frac{t^n}{n} \quad -1 \leq t < 1\end{aligned}$$

令 $b \rightarrow 1^-$

$$\varphi(1) = \int_0^1 \frac{-\ln(1-x)}{x} dx = \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

4. 当 $\alpha = 0$ 时, $\checkmark$

$$\begin{aligned}\text{当 } \alpha > 0 \text{ 时, } x \rightarrow 1^+ \text{ 时, } f(x, \alpha) &\sim \frac{\arctan \alpha}{\sqrt{2}(\alpha-1)} \\ x \rightarrow +\infty \text{ 时, } f(x, \alpha) &\sim \frac{\frac{\pi}{2}}{x^3}\end{aligned}$$

$\int_1^{+\infty} f(x, \alpha) dx$ 收敛

$$\frac{\partial f(x, \alpha)}{\partial \alpha} = \frac{1}{x\sqrt{x^2-1} [1+(\alpha x)^2]}$$

当 $\alpha \geq 0, x \geq 1$ 时.

$$0 \leq \frac{1}{x\sqrt{x^2-1} [1+(\alpha x)^2]} \leq \frac{1}{x\sqrt{x^2-1}}$$

由 Weierstrass 判别法.

$\Rightarrow$ 含参变量广义积分 $\int_1^{+\infty} \frac{\partial f(x, \alpha)}{\partial \alpha} dx$ 在 $\alpha \geq 0$ 上一致收敛.

$$\varphi'(\alpha) = \int_1^{+\infty} \frac{1}{x\sqrt{x^2-1} [1+(\alpha x)^2]} dx$$

$$\stackrel{x=\sec \theta}{=} \int_0^{\frac{\pi}{2}} \frac{1}{1+(\alpha \sec \theta)^2} d\theta$$

$$= \int_0^{\frac{\pi}{2}} \frac{1}{1+\alpha^2 + \alpha^2 \tan^2 \theta} d\theta$$

$$= \int_0^{+\infty} \frac{1}{(1+\alpha^2 + \alpha^2 t^2)(1+t^2)} dt$$

$$\varphi'(0) = \frac{\pi}{2} \quad \alpha > 0 \quad \text{令 } \beta = \frac{\sqrt{\alpha^2+1}}{\alpha} > 1$$

$$\varphi'(\alpha) = \frac{1}{\alpha^2} \int_0^{+\infty} \frac{dt}{(t^2+1)(t^2+\beta^2)}$$

$$= \frac{1}{\alpha^2} \frac{1}{\beta^2-1} \int_0^{+\infty} \left(\frac{1}{t^2+1} - \frac{1}{t^2+\beta^2} \right) dt$$

$$= \frac{1}{\alpha^2} \frac{1}{\beta^2-1} \left(\frac{\pi}{2} - \frac{\pi}{2} \frac{1}{\beta} \right)$$

$$= \frac{\pi}{2} \frac{1}{\alpha^2 \beta (\beta+1)}$$

$$\varphi(\alpha) = \frac{\pi}{2} (\alpha - \sqrt{1+\alpha^2} + C)$$

由 $\varphi(0) = 0 \Rightarrow C = 1$.

$$\text{所以 } \varphi(\alpha) = \frac{\pi}{2} (\alpha - \sqrt{1+\alpha^2} + 1) \quad \alpha > 0$$

$$\text{同理, } \alpha < 0 \text{ 时, } \varphi(\alpha) = \frac{\pi}{2} (\alpha + \sqrt{1+\alpha^2} - 1)$$

$$\Rightarrow \varphi(\alpha) = \frac{\pi}{2} (|\alpha| + 1 - \sqrt{1+\alpha^2}) \operatorname{sgn} \alpha.$$

证: (1) 方法一: 可以证明, 当 $x > 0$ 时, $\ln(1+x) < \sqrt{x}$

$$\Leftrightarrow \text{当 } y > 0 \text{ 时, } \ln(1+y^2) < y$$

$$\text{令 } f(y) = y - \ln(1+y^2) \quad f(0) = 0$$

$$\text{当 } y > 0 \text{ 时, } f'(y) = 1 - \frac{2y}{1+y^2} = \frac{(y-1)^2}{1+y^2} \begin{cases} > 0 & y \neq 1 \\ = 0 & y = 1 \end{cases}$$

当 $0 \leq t \leq T, x > 0$ 时,

$$0 \leq \frac{\ln(1+tx)}{1+x^2} \leq \frac{\sqrt{T}\sqrt{x}}{1+x^2}$$

$$\int_0^{+\infty} \frac{\sqrt{T}\sqrt{x}}{1+x^2} dx \text{ 收敛. 由 Weierstrass 判别法}$$

$$\Rightarrow \int_0^{+\infty} \frac{\ln(1+tx)}{1+x^2} dx \text{ 关于 } t \text{ 在 } [0, T] \text{ 上一致收敛}$$

方法二: 当 $0 \leq t \leq T, x \geq 0$ 时.

$$0 \leq \frac{\ln(1+tx)}{1+x^2} \leq \frac{\ln(1+Tx)}{1+x^2}$$

$$\int_0^{+\infty} \frac{\ln(1+Tx)}{1+x^2} dx \text{ 收敛. 由 Weierstrass 判别法}$$

$$\Rightarrow \int_0^{+\infty} \frac{\ln(1+tx)}{1+x^2} dx \text{ 关于 } t \text{ 在 } [0, T] \text{ 上一致收敛}$$

(2) 当 $t > 0$ 时,

$$\varphi\left(\frac{1}{t}\right) = \int_0^{+\infty} \frac{\ln(1+\frac{1}{t}x)}{1+x^2} dx$$

$$\stackrel{x=\frac{1}{u}}{=} \int_{+\infty}^0 \frac{\ln(1+\frac{1}{tu})}{1+\frac{1}{u^2}} \left(-\frac{1}{u^2}\right) du$$

$$= \int_0^{+\infty} \frac{\ln(1+tu) - \ln t - \ln u}{1+u^2} du$$

$$= \int_0^{+\infty} \frac{\ln(1+tu)}{1+u^2} du - \frac{\pi}{2} \ln t - 0$$

$$= \varphi(t) - \frac{\pi}{2} \ln t$$

$$\int_0^{+\infty} \frac{\ln u}{1+u^2} du \stackrel{u=\frac{1}{x}}{=} \int_{+\infty}^0 \frac{-\ln x}{1+\frac{1}{x^2}} \left(-\frac{1}{x^2}\right) dx$$

$$= - \int_0^{+\infty} \frac{\ln x}{1+x^2} dx$$

$$\Rightarrow \int_0^{+\infty} \frac{\ln u}{1+u^2} du = 0$$